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Masters Thesis

# Correlations in Non-Markovian Open Quantum System Dynamics

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# Introduction

In quantum mechanics, the assumption of an isolated system is often impractical when describing real-world phenomena. While idealized models offer significant insights into fundamental principles, they fail to capture the full complexity of systems interacting with their environment. In reality, many quantum systems are coupled to their surroundings, leading to dynamics that deviate from unitary evolution [1–3].

Such systems, termed open quantum systems (OQS), present significant modelling challenges. The continuous exchange of energy and information with the environment introduces decoherence, dissipation, and noise. These are inherently non-unitary processes that cannot be described by the Schrödinger equation alone. The fundamental difficulty lies in characterizing the environmental influence. With infinite degrees of freedom, fully simulating the bath dynamics becomes computationally intractable. Consequently, OQS theory focuses on deriving effective descriptions that capture environmental effects without explicitly tracking all bath modes, instead working within reduced Hilbert spaces [4].

The Markov approximation has emerged as one of the most successful frameworks to describe OQS dynamics [5]. This approach assumes memoryless evolution, where the system's future depends only on its present state, not its history. Physically, this corresponds to a regime where information flows unidirectionally from system to bath, with no backflow of correlations. Additionally, the approximation requires weak system-bath coupling to ensure the bath remains essentially unperturbed. However, as experimental capabilities advance, enabling the creation and control of increasingly complex quantum systems, these assumptions often break down. In such regimes, non-Markovian effects become prominent, characterized by information backflow from the environment to the system and the persistence of system-bath correlations [6].

The emergence of correlations between a quantum system and its environment lies at the core of the challenges associated with non-Markovian dynamics. At the onset of interaction, the system and bath are typically assumed to be uncorrelated, and their joint state is well described by a product state. However, this assumption quickly breaks down as the system evolves. Entanglement and correlations naturally build up between the system and its environment, significantly altering the dynamics [7, 8]. In the weak coupling regime, these correlations remain sufficiently small such that their effects can often be neglected. This underpins the success of the Markov approximation, wherein the environment is assumed to have no memory of past interactions and system bath coupling remains weak. However, as the system-bath coupling strength increases or the environment exhibits spectral structure, such as sharp features or band edges, these correlations become non-negligible. In such regimes, the environment can

retain and later return information about the system's past, leading to memory effects characteristic of non-Markovian behaviour. These memory effects manifest in a range of experimentally observable phenomena, including coherence revivals, non-exponential relaxation, and transient reversals in the direction of information flow between system and environment [9, 10]. Accurately capturing such dynamics requires theoretical and numerical approaches that go beyond the Markovian limit and can account for the time-dependent build-up and feedback of system-environment correlations.

Importantly, system-environment correlations are not only a challenge to be mitigated, but also a resource to be exploited for quantum control [11]. Exploiting these correlations can enhance the controllability of quantum systems, enabling more efficient state preparation, coherence preservation, or entanglement generation than would be possible under Markovian dynamics alone. By deliberately shaping the environment or engineering its interaction with the system, it is possible to steer the dynamics in a desired direction, even in the presence of strong coupling or non-Markovian memory effects. This perspective highlights the dual role of the environment, not only as a source of noise, but also as a potential partner in coherent quantum control.

These insights have significantly reshaped the role of the environment in quantum dynamics, from a passive source of decoherence to an active participant that can be shaped and even harnessed for control. Exploiting system–environment correlations opens up new possibilities for quantum technologies. However, accurately describing these correlations requires modelling frameworks that extend beyond the traditional Markovian approximation. The increased dependence on environmental dynamics is a key reason why non-Markovian systems are so challenging to simulate. The central difficulty lies in incorporating memory effects—arising from system–bath correlations without explicitly modelling the full, infinite-dimensional environment. A comprehensive overview of this problem is provided in [4].

Perturbative techniques, such as those leading to master equations, eliminate the environmental degrees of freedom analytically before time evolution, then adding on dissipation and noise effects separately. While computationally efficient, this procedure can lose information about essential correlations and memory effects. To address this limitation, non-Markovian master equations have been developed, which incorporate memory kernels or time-dependent generators [12, 13]. While accurate, these equations are often quite complex and difficult to solve. In contrast, a more intuitive and physically transparent approach is to model a section of the environment (the effective modes) explicitly alongside the system. Propagating both the system and truncated bath as a closed system. However, these methods are complicated by the infinite nature of environmental baths and how one can accurately truncate it.

This thesis investigates two such methods based on the effective modes approach to simulating open quantum systems. Both methods approximate the environment using a finite, discrete set of modes, thereby enabling the joint propagation of the system and a truncated bath. This approach retains key system–environment correlations and allows the simulation to capture non-Markovian features such as information backflow and memory-dependent dynamics, while remaining computationally tractable.

This thesis is structures as follows: firstly a quick introduction is given to open quantum systems and non-Markovian dynamics. The second chapter introduces the method of exact mapping an OQS to a chain. This chapter outlines the method, discussing the idea of self convergence Chapter two introduces the method of exactly mapping an OQS to a one dimensional chain. This chapter outlines the theoretical basis of the method, discusses the idea of self convergence self-convergence, the systematic improvement of accuracy as truncated bath size increases, and explains how matrix product state techniques can be used to propagate the dynamics efficiently. The spin-boson model serves as the primary test case [14], this model provides as comparison point for approximation dynamics used throughout the thesis. The convergence properties of this model are then discussed. This reveals the specific parameter regimes where finite-size effects are negligible, establishing practical guidelines for implementation. Chapter 3 introduces the second effective mode approach, known as the surrogate Hamiltonian method. Unlike chain mapping, this method truncates the environment by directly sampling a subset of bath modes, typically those near resonance with the system [15]. The underlying principles of the method are discussed, along with the criteria required for self-convergence. As before, the spin-boson model is approximated, and the impact of both bath size and sampling strategy on convergence is explored. The fourth chapter compares the dynamics of these two approximation methods against the Markov approximation [5]. These three methods are compared first in the weak coupling regime where the Markov approximation is valid. Allowing for direct comparison of the accuracy of both approximation methods. This comparison is then extended to the strong coupling regime where the Markov approximation is no longer valid. This regime provides a starting point to discuss how each effective mode method treats non-Markovianity and see how the methods differ in their treatment of correlations. Finally, Chapter 5 summarises the main findings and outlines potential directions for future research.

# Chapter 1

# Open Quantum Systems and non-Markovian Dynamics

This chapter provides an overview of open quantum system simulation methods and explains why non-Markovian dynamics necessitate developing approaches that go beyond the Markovian limit.

Open quantum systems (OQS) evolve under the influence of both their intrinsic dynamics and environmental interactions. The total Hamiltonian takes the form

$$H = H_S + H_B + H_I, \tag{1.1}$$

where $H_S$ governs the system, $H_B$ describes the bath, and $H_I$ couples them. Directly propagating the full Hamiltonian is computationally intractable. The bath's infinite-dimensional nature makes it impossible to fully propagate in finite time. Therefore, in order to simulate the dynamics of an OQS, one must first truncate or remove the bath in some way that retains the effects of the bath on the system.

Two fundamentally different regimes emerge, distinguished by how information flows between system and environment. In Markovian dynamics, information flows unidirectionally from system to bath, never returning. Non-Markovian dynamics exhibit backflow of information, with memory playing a crucial role.

## 1.1 The Markov Approximation

The Markov approximation [5] is a widely used framework for describing OQS dynamics. This approximation assumes that the bath recovers instantly from interactions with the system, leading to memoryless evolution with a continuous flow of information from the system to the environment. The method works by first partially tracing out the environmental degrees of freedom and then evolving the system using a dynamical map or master equations. More details about the derivation of the Markov approximation can be found in [1].

The Markov approximation relies on two key assumptions. The first being weak system-bath coupling, such that the coupling energy is much smaller than bath energy scales. This ensures the bath remains close to thermal equilibrium and system-bath entanglement remains negligible. In the interaction picture, this allows the total

density matrix to be approximated as

$$\rho_{SB}(t) \approx \rho_S(t) \otimes \rho_B. \tag{1.2}$$

Secondly, it is assumed that bath correlations decay much faster than the system's relaxation time. This eliminates memory effects, leading to Markovian dynamics where the future evolution depends only on the current system state, not on the history of system-environment interactions. These approximations enable the derivation of a master equation that describes only the system evolution while incorporating environmental effects through dissipative terms. The system dynamics are obtained by tracing out the bath degrees of freedom

$$\rho_S(t) = \mathrm{Tr}_B[\rho_{SB}(t)]. \tag{1.3}$$

Systems that abide by these assumptions are said to be in the Markovian limit. The dynamics of OQS in this limit can be described by the master equation [5]

$$\frac{d\rho_S}{dt} = \mathcal{L}(\rho_S) = -i[H_S, \rho_S] + \sum_k \gamma_k [L_k \rho_S L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho_S\}] \tag{1.4}$$

where $H_S$ is the system Hamiltonian and $\{L_k\}$ are the Lindblad jump operators. The first term describes the unitary evolution of the system and the second term describes the dissipation caused by the environment interacting with the system. The Lindblad jump operators $L_k$ and decay rates $\gamma_k$ are determined by the operator acting on the system in the interaction Hamiltonian $\hat{A}$, the bath's spectral density $J(\omega_{nm})$ and the system Hamiltonian's eigenvectors

$$\gamma_{nm} = |\langle n|\hat{A}|m\rangle|^2 J(\omega_{nm}), \tag{1.5}$$

$$L_{nm} = \sqrt{\gamma_{nm}}|n\rangle\langle m|. \tag{1.6}$$

This approximation works well for many systems, but its assumptions of Markovian dynamics and weak coupling make it unsuitable for strongly coupled or non-Markovian systems.

## 1.2 Beyond the Markovian Limit

The limitations of the Markov approximation have motivated the development of alternative approaches capable of handling non-Markovian dynamics and strong system-bath coupling. Non-Markovian behaviour emerges when the standard assumptions break down: the bath retains memory of past interactions, correlations persist between system and environment, and information flows bidirectionally rather than simply dissipating away [16].

Non-Markovian dynamics have been experimentally observed across diverse physical platforms, from photon-atom interactions in cavity QED [17, 18] to charge and flux qubits in solid-state devices [19, 20], and even in biological systems where quantum coherence assists energy transfer [21, 22]. This ubiquity underscores the need for theoretical methods that go beyond the Markovian limit.

The comprehensive review by de Vega and Alonso [4] categorizes existing non-Markovian approaches. Some approaches such as master equations, Heisenberg equations and

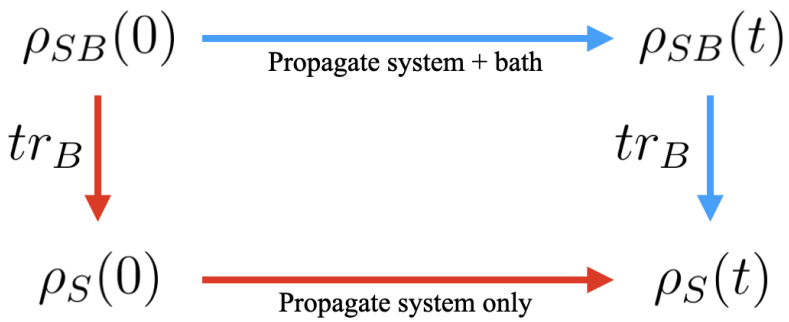



Figure 1.1: Comparison of two approaches to open quantum system simulation. Red arrows show approaches like the perturbative master equation where the environmental bath is traced out initially and only system dynamics are propagated. This contrasts with methods detailed in this thesis that follow the blue arrows, where the full system-environment evolution is approximated to enable propagation of both system and bath, with system dynamics recovered by tracing out the bath post-propagation. Adapted from [16].

stochastic methods are based on propagate the system only, adding on terms that account for the environmental influence. Whilst computationally effective in the Markovian limit, these approaches are challenged by the inherent non locality of non-Markovian dynamics. Other techniques rely on an effective mode approach where some form of the full system-bath dynamics are propagated. But these approaches are computationally expensive.
This thesis explores the second strategy. Rather than eliminating bath degrees of freedom through before propagation, a finite but explicit representation of the bath is retained that evolves alongside the system. Only after the full propagation is complete is the bath trace out to obtain the reduced system dynamics. As illustrated in Figure 1.1, instead of simplifying first and propagating later (red), these methods propagate the combined system-bath entity and extract the system information afterwards (blue). This strategy naturally captures memory effects and strong correlations that arise during the evolution, at the cost of requiring efficient methods to handle the enlarged Hilbert space.

## 1.3 State Space Volume as a Measure of non-Markovianity

The volume occupied by an ensemble of quantum states in Hilbert space provides a geometric framework to characterize the system's accessible information [23].

For an ensemble of pure states $|\psi_i\rangle$, this volume evolves under the influence of the system's dynamics. In the Markovian limit the state space volume contracts monotonically over time. This is due to the constant loss of information from the system into the environment. However, outside of the Markovian limit information can travel from the environment into the system. This backflow of information is characterised as an increase in the state space volume.

This distinction is illustrated in Figure 1.2: panel (a) shows a monotonic, exponential decay of the state space volume typical of Markovian behaviour, whereas panel

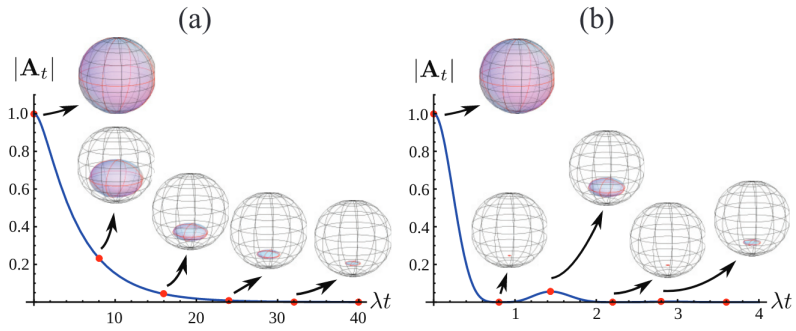



Figure 1.2: Evolution of the volume of accessible quantum states $|A_t|$ for (a) Markovian and (b) non-Markovian dynamics. In the Markovian case, the volume decreases steadily, reflecting a continuous loss of information to the environment. In contrast, the non-Markovian case shows temporary increases in volume, indicating information backflow from the environment to the system. These changes illustrate how the structure of the system dynamics changes due to the flow of information and the degree of memory in the system's evolution. Taken from [23]

(b) displays a clear non-monotonic revival in volume, indicating information backflow.

While the absence of volume increase does not definitively confirm Markovianity, any non-monotonic growth in the state space volume provides a sufficient indicator of non-Markovianity [23].

# Chapter 2

# The Chain Mapping Method

An approach to approximating open quantum systems, proposed by Chin *et al.* [14], employs an exact unitary transformation to map a linearly coupled open system onto a semi-infinite one-dimensional chain. This method, which we refer to as the chain mapping method throughout this thesis, leverages orthogonal polynomial properties and the separability of topological space to transform a system coupled to many bath modes into a chain with only nearest-neighbour interactions. The resulting one-dimensional structure proves particularly advantageous for numerical simulation, as it naturally suits propagation using matrix product state (MPS) techniques.

This chapter presents an analysis of the chain mapping method. It begins by examining the mathematical construction of the transformation operator and its physical interpretation, drawing primarily from the foundational works [14, 24–26]. The next section demonstrates the method's application to the spin-boson model as seen in [14], providing concrete examples of the transformation procedure and resulting chain parameters. The chapter concludes with a critical assessment of the method's convergence properties and how numerical parameters such as bond dimension effect convergence.

## 2.1 The idea of Chain Mapping

The chain mapping method exploits a fundamental property of open quantum systems, namely that the system-bath interaction depends entirely on the bath's spectral density rather than the specific arrangement of bath modes. This insight enables a powerful reorganization where the continuous bath modes are mapped onto a semi infinite one-dimensional chain, transforming an exponentially complex problem into one with linear scaling. The resulting chain can be truncated to a finite length while still capturing the essential bath dynamics, making it computationally feasible to propagate both system and truncated bath together. This explicit treatment of bath evolution proves crucial for capturing non-Markovian effects, as memory and correlation build-up are naturally included. Unlike perturbative methods where bath degrees of freedom are eliminated from the outset, part of the full system-bath evolution is retained throughout the calculation in chain mapping. The bath dynamics are propagated explicitly alongside the system, and the build-up of correlations and memory effects that characterize non-Markovian behaviour are captured. After the evolution is computed, the bath dynamics are traced out to obtain the original system dynamics. This approach does not require assumptions of Markovianity or weak coupling, as

information backflow and strong coupling effects are naturally incorporated through the inclusion of bath dynamics in the calculations.

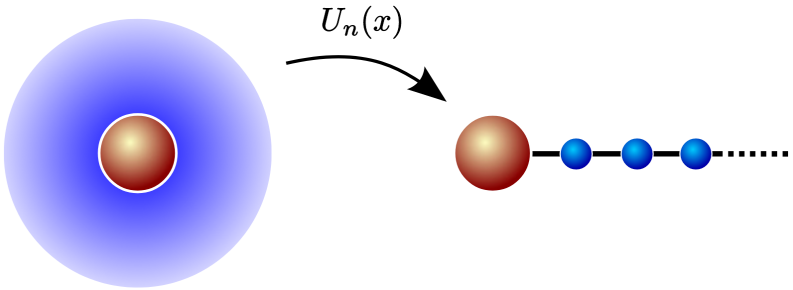



Figure 2.1: Illustrated representation of a system being mapped onto a one dimensional chain. The arrow represents the effect of the transformation operator $U_n(x)$ which exactly maps the original system interacting with a continuous bath (left) to a semi infinite one dimensional chain (right). Taken from [14].

An infinite environment coupled to a system can be described as a continuum of bosonic or fermionic modes. The dynamics of this OQS are generated by the Hamiltonian

$$\hat{H} = \hat{H}_{\text{Sys}} + \hat{H}_{\text{Env}} + \hat{H}_{\text{Int}} = \hat{H}_{\text{Sys}} + \int_0^\infty dx\, g(x)\hat{a}_x^\dagger \hat{a}_x + \hat{A}_s \int_0^\infty dx\, h(x)(\hat{a}_x + \hat{a}_x^\dagger), \quad (2.1)$$

where $\hat{A}_s$ is a coupling operator acting on the system, $g(x)$ is the dispersion relation, $h(x)$ is the coupling function and $\hat{a}_x^\dagger$ and $\hat{a}_x$ are the creation and annihilation operators for each mode $x$. The spectral density function is defined

$$J(\omega) = \pi h^2 \left[g^{-1}(\omega)\right] \rho(\omega) \quad (2.2)$$

where $\rho(\omega) = \frac{dg^{-1}(\omega)}{d\omega}$ is the density of states in the bath. Given the system is coupled linearly to some environment, the dynamics induced on the system by the bath are determined entirely by the spectral density function $J(\omega)$ [3, 27]. This means that the functions $g(x)$ and $h(x)$ are not uniquely defined as long as equation (2.2) is satisfied.

To simplify the transformation operator $U_n(x)$, $h(x)$ is chosen such that $g(x)$ is linear: $g(x) = gx$. This selection allows the transformation operator definition to be defined as,

$$U_n(x) = h(x)\tilde{p}_n(x), \quad (2.3)$$

where $\tilde{p}_n(x)$ are a set of orthonormal polynomials with respect to the measure $d\mu(x) = h(x)^2 dx$. This measure is chosen so that the set $\tilde{p}_n(x)$ is orthogonal with respect to the spectral density function $J(\omega) \propto h^2(\omega)$. The unitarity of $U_n$ is ensured by the orthogonality condition

$$\int_0^\infty dx\, U_n(x) U_m^*(x) = \delta_{nm}. \quad (2.4)$$

These operators $Un(x)$ form an orthonormal basis for the environment's Hilbert space, allowing the original continuous-mode operators $\hat{a}_x^\dagger$ and $\hat{a}_x$ to be expanded in terms of a discrete set of modes associated with the chain. This basis transformation enables

the definition of a new set of bosonic (or fermionic) operators corresponding to the chain sites. This transformed set of creation and annihilation operators $b_n^\dagger$ and $b_n$ are defined

$$b_n^\dagger = \int_0^\infty dx\, U_n(x) a_x^\dagger, \quad \text{and} \quad b_n = \int_0^\infty dx\, U_n(x) a_x. \tag{2.5}$$

These operators satisfy commutation relations $[b_n, b_n^\dagger] = \delta_{nm}$ for bosons and anti-commutation relations $\{b_n, b_n^\dagger\} = \delta_{nm}$ for fermions.

The transformation operator is applied to the Hamiltonian in equation (2.1) by expressing the creation and annihilation operators in terms of their transformed counterparts $b_n^\dagger$ and $b_n$ and interacting the system with only the first chain mode ($n = 0$). This results in the following mapped Hamiltonian,

$$H' = H_{\text{Sys}} + c_0\ \hat{A}_s \left(b_0 + b_0^\dagger\right) + \sum_{n=0}^{\infty} \left(\omega_n b_n^\dagger b_n + t_n b_{n+1}^\dagger b_n + t_n b_n^\dagger b_{n+1}\right), \tag{2.6}$$

where $c_0 = \sqrt{\int_0^\infty J(\omega) d\omega}$ and, $\omega_n$ are the frequencies of each chain mode and $t_n$ are the tunnelling amplitudes along the chain links. They are chosen using the recursion coefficients of the set of orthonormal polynomials used in the transformation operator (Eq. 2.3). The derivation of these values is detailed in [14].

In practice, the infinite chain must be truncated for numerical simulation in finite time. The sum in Eq. 2.6 is therefore limited to a finite chain length $N$, chosen such that boundary effects do not influence the system dynamics within the desired simulation time. It is also important to note that this transformation does not map each bath mode from the original system one-to-one. Instead, it constructs a new set of chain modes, ordered such that the modes at the beginning of the chain capture the most significant correlations with the system. This prioritization allows a truncated chain to retain the dominant influences of the bath, rather than preserving specific bath modes as in the surrogate Hamiltonian approach.

### 2.1.1 Self Convergence

The convergence time marks when boundary effects from the finite chain begin to corrupt the dynamics. Before this time, the truncated chain accurately reproduces the infinite bath behaviour. After this time, information and correlations that have propagated to the chain's end reflect back toward the system, causing unphysical deviations from the true dynamics.

These boundary reflections arise because information and correlations that would dissipate forever into an infinite bath instead encounter the artificial end of the truncated chain. The recurrences have no physical counterpart in the original system and can superficially resemble non-Markovian dynamics. However, they are an artifact of the finite bath size and occur regardless of the system's actual non-Markovian behaviour. In a truly infinite bath, these recurrences would simply propagate away from the system and never return.

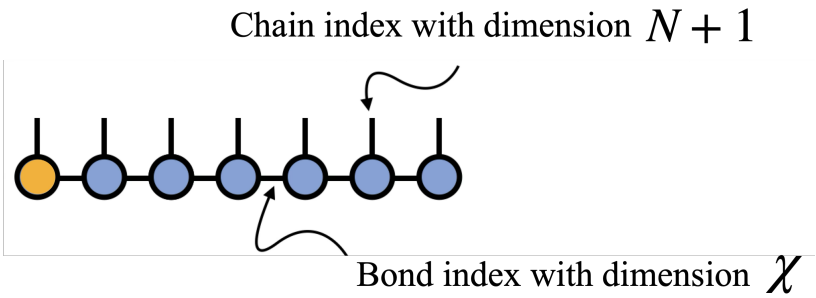



Figure 2.2: Illustrated picture of a MPS of a chain mapped OQS with bath length $N$. The system (orange) is connected to each chain mode (blue) with a link. These links have a bond dimension $\chi$ which decides the maximum amount of entanglement between chain links. Adapted from [30]

Self-convergence is achieved when the chain length $N$ is sufficient to prevent boundary effects within the simulation time frame. This condition is verified by increasing $N$ and confirming that the dynamics remain unchanged over the time interval of interest. Once self-convergence is established, the finite chain provides an accurate representation of the infinite bath dynamics for that specific duration.

### 2.1.2 Propagating with Matrix Product State methods

One main advantage of this approximation method is that its structure matches that of a Matrix Product State (MPS), where each site is connected to its nearest neighbour by some bond dimension $\chi$ [28, 29]. Due to its one dimensional tensor network structure, the Hilbert space of a MPS increases linearly with the number of sites.

A MPS is one of the most well understood tensor networks. Due to its simplicity, a significant amount of efficient algorithms have been developed, in particular for the simulation of one dimensional quantum many body dynamics [31]. While early approaches like time-dependent DMRG (tDMRG)[32] adapted the popular time-independent DMRG sweeping algorithm [33] to time-dependent dynamics, similar approaches like Time-Evolving Block Decimation (TEBD) [34, 35] is more commonly used due to its inherent unitarity and systematic error control. In this thesis we employ TEBD as our propagation method.

TEBD uses Trotter decomposition of the Hamiltonian to evolve a Matrix Product State (MPS) in time. This decomposition splits the time evolution operator into local two-site operations, which can be applied sequentially in a way that approximately preserves the MPS structure, enabling efficient simulation of time-dependent evolution. To apply these local gates, the Hamiltonian is typically split into terms acting on even and odd bonds. Two-site unitary gates are then applied to each bond in a sequence. However, the application of these gates increases the bond dimension of the MPS. To prevent the bond dimension from growing uncontrollably, it is truncated after each step.

The TeNPy library [36] used in this thesis to implement TEBD handles this truncation using two parameters: the maximum bond dimension $\chi_{\max}$, which sets an upper limit

on the entanglement that can be represented between sites, and a Schmidt value truncation threshold, which sets a lower bound for the singular values to be retained. After applying a gate, a Schmidt decomposition is performed. If the number of Schmidt values above the threshold is less than $\chi_{\max}$, all are kept. If more than $\chi_{\max}$ survive the threshold, only the largest $\chi_{\max}$ values are retained. This ensures the bond dimension is controlled without discarding significant contributions to the entanglement structure.

While TEBD's efficiency comes from exploiting the one-dimensional structure of the MPS, this naturally limits its direct application to only one-dimensional Hamiltonians. However, this apparent limitation is precisely why the mapping to a one-dimensional chain presented in this chapter is so powerful. By mapping a system coupled to many bath modes onto a one-dimensional chain, TEBD's computational efficiency can be leveraged for problems that would otherwise require exponentially scaling methods. This combination of chain mapping with TEBD has opened up the study of strongly coupled, non-Markovian open quantum systems to numerically exact treatment, as we demonstrate in the following sections.

## 2.2 Approximating the Spin-Boson model

To illustrate the chain mapping method, it is applied to the spin-boson model as presented in Ref. [14]. In this example, a two-level spin system interacts with a bosonic bath, serving as a standard test case for open quantum systems.

To facilitate the chain mapping, the dimensionless frequency variable $x = \omega/\omega_c$ is used. For a hard cut-off bath, $x \in [0, 1]$, while for an exponential cut-off, $x \in [0, \infty)$. Under this transformation, the Hamiltonian becomes

$$\hat{H} = \omega_0 \hat{\sigma_z} + \int_0^{x_{\max}} dx\, g(x) \hat{a}_x^\dagger \hat{a}_x + \frac{1}{2}\hat{\sigma}_x \int_0^{x_{\max}} dx\, h(x)(\hat{a}_x + \hat{a}_x^\dagger), \tag{2.7}$$

where $\hat{\sigma}_z$ is the Pauli z operator acting on the spin system, $\hat{a}_x$ and $\hat{a}_x^\dagger$ are the bosonic creation and annihilation operators for the $x$th bath mode, $g(x)$ is the dispersion relation, and $h(x)$ is the system-bath coupling function.

As mentioned above, the spectral density fully characterises the bath's influence on the system dynamics. Below two bath structures are presented.

**Hard Cut-off Bath**

For a bath with sharp frequency cut-off at $\omega_c$ the spectral density is defined

$$J^{hard}(\omega) = 2\alpha\pi\omega_c^{1-s}\omega^s\Theta(\omega_c - \omega), \tag{2.8}$$

where $\alpha$ is the dimensionless coupling strength, $\Theta(\omega - \omega_c)$ is the Heaviside step function and $s$ controls the frequency dependence of the system-bath interaction ($s = 1$ for ohmic systems).

From equation (2.2), the dispersion relation and coupling function are defined

$$g(x) = \omega_c x, \tag{2.9}$$

$$h^{hard}(x) = \sqrt{2\alpha}\omega_c x^{s/2}. \tag{2.10}$$

Therefore, the transformation operator for this system is

$$U_n^{hard}(x) = x^{s/2}\tilde{P}_n^{(0,s)}(x), \tag{2.11}$$

where $\tilde{P}_n^{(0,s)}(x)$ is a normalized shifted Jacobi polynomial [37]. This transformation yields a chain with site frequencies

$$\omega_n^{hard} = \frac{\omega_c}{2}\left(1 + \frac{s^2}{(s+2n)(2+s+2n)}\right), \tag{2.12}$$

and the nearest-neighbour tunnelling amplitudes

$$t_n^{hard} = \frac{\omega_c(1+n)(1+s+n)}{(s+2+2n)(3+s+2n)}\sqrt{\frac{3+s+2n}{1+s+2n}}. \tag{2.13}$$

To simulate the bath with a finite number of modes, the chain is truncated at site index $n = N$, and the above expressions are evaluated only up to this point.

**Exponential Cut-off bath**

The second bath structure presented is that with an exponential cutoff. Here the bath cut-off is smooth rather than a sharp boundary. The spectral density in this case is

$$J^{exp}(\omega) = 2\pi\alpha\omega_c^{1-s}\omega^s e^{-\omega/\omega_c}. \tag{2.14}$$

Following the same procedure as before with $x_{\max} = \infty$, the coupling function becomes

$$h^{exp}(x) = \sqrt{2\alpha}\omega_c x^{s/2}e^{-s/2}. \tag{2.15}$$

With $g(x) = \omega_c x$ remaining the same as above.

Instead of Jacobi polynomials, this bath uses associated Laguerre polynomials. This results in the transformation operator

$$U_n^{exp}(x) = x^{s/2}\tilde{L}_n^s(x), \tag{2.16}$$

with $\tilde{L}_n^s(x)$ being a normalised associated Laguerre polynomial [37]. The resulting frequencies of the exponential cut-off bath are

$$\omega_n^{exp} = \omega_c(2n+1+s), \tag{2.17}$$

and the corresponding tunnelling amplitudes are

$$t_n^{exp} = \omega_c\sqrt{(n+1)(n+s+1)}. \tag{2.18}$$

Compared to the hard cut-off bath, these chain parameters grow linearly with time and are unbounded, as one would expect in an exponential cut-off bath.

Substituting either the hard or exponential cut-off frequencies and tunnelling amplitudes into equation (2.6) along with the transformed creation and annihilation operators ($\hat{b}^\dagger$,$\hat{b}$) results in the one dimensional mapped hamiltonian for this spin boson system.

## 2.3 Convergence Analysis

The accuracy of the chain-mapped simulation depends primarily on two parameters: the chain length $N$ and the maximum bond dimension $\chi_{\max}$. The chain length determines how long the approximation remains valid before boundary effects appear, while the bond dimension controls how much entanglement the system can represent [38]. In this section, the influence of both parameters on the numerical convergence of the spin-boson model is analysed across weak and strong coupling regimes.

### 2.3.1 Convergence with Chain Length

An infinitely long chain cannot be propagated in finite time. Therefore, the chain must be cut to some finite length $N$. For finite $N$, information and correlations that propagate along the chain will eventually reach the end and reflect back, causing recurrences in the system dynamics. These recurrences cause the approximation to diverge from the true dynamics.

Using the spin-boson model described by Eq (2.7), we begin with a spin in its up state $|\uparrow\rangle$ coupled to a bath in the ground state $|0000...\rangle$. Since the initial state of the system is an eigenstate of $\sigma_z$, $\langle\sigma_x\rangle = \langle\sigma_y\rangle = 0$ throughout the evolution. Allowing the Bloch sphere dynamics of the system to be described by only $\langle\sigma_z\rangle$. For the simulations below, the system frequency $\omega_0 = 1$, the cut-off frequency $\omega_c = 5$, and an Ohmic spectral density $s = 1$ are used.

Figure 2.3 shows $\langle\sigma_z\rangle$ for different chain lengths $N$ in very weak ($\alpha = 10^{-3}$), weak ($\alpha = 10^{-1}$), and strong ($\alpha = 1$) coupling regimes. The finite chain truncation introduces artificial recurrences caused by the boundary conditions that alter the long-time dynamics. While the infinite bath provides a true dissipative environment where information and correlations propagate away irreversibly, finite chains force them to reflect at the boundary and return to the system, causing deviation from true dynamics.

It follows that shorter chains deviate from the true dynamics earlier than longer chains. The time it takes for an approximation, with chain length $N$, to diverge from the dynamics of approximations with longer chains is called the convergence time. During this period, information sent from the system into the bath has not yet reached the chain boundary, making the dynamics indistinguishable from the infinite case. Once past the convergence time, boundary reflections corrupt the dynamics.

Looking again at Figure 2.3, convergence time exhibits no dependence on the coupling strength $\alpha$. While stronger coupling increases the magnitude of information and correlations propagating into the bath, it does not affect their velocity. The speed at which information and correlations travel through the bath remains constant. This is further seen in the chain parameters in equations (2.12) and (2.13), as they depend only on the $s$ and cut-off frequency $\omega_c$, not on $\alpha$. This finite propagation speed can be understood through the Lieb-Robinson bound, which sets a maximum velocity for information propagation in local Hamiltonians [39].

Figure 2.4 quantifies this behaviour by plotting convergence time versus chain length. The observed linear relationship, confirms the existence of a constant velocity for in-

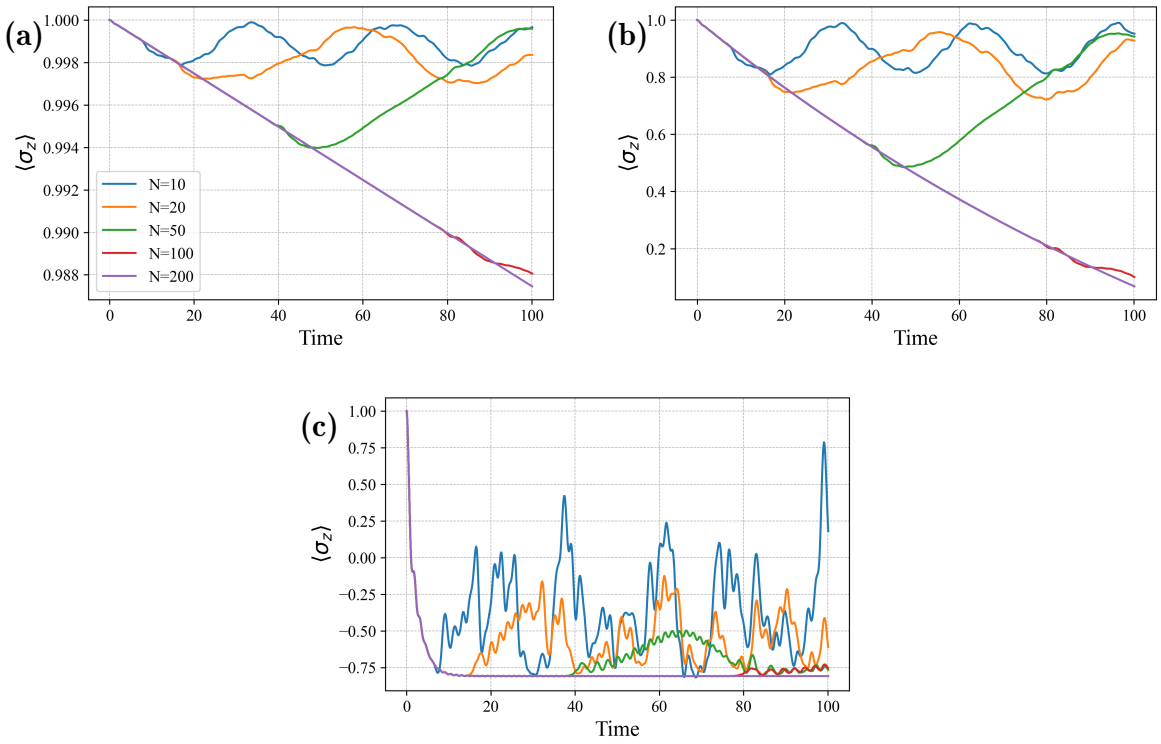



Figure 2.3: Effects of chain length on approximation convergence for different coupling regimes, (a) very weak coupling $\alpha = 10^{-5}$ (b) weak coupling $\alpha = 0.001$ and (c) strong coupling $\alpha = 0.1$. All coupling strengths show the same convergence time for the same chain length $N$. This indicates that the speed at which information and correlations travel down the chain is independent of coupling strength. Parameters $\chi_{\max} = 60$.

formation and correlations moving in the chain. This linear scaling also provides a practical prescription for choosing chain lengths. To accurately capture dynamics up to time $T$, one requires a chain length $N$ with a convergence time greater than $T$. This allows one to control how long the approximation needs to be valid for, potentially reducing the computational time needed for short time dynamics.

Additionally, due to the linear scaling of the Hilbert space with the number of modes $N$, and the computational efficiency of matrix product state (MPS) techniques, the chain mapping approximation can be extended to significantly larger chain lengths than many other effective mode methods. This scalability is particularly advantageous in regimes where long-range memory effects or slow bath decay times require a more extended environmental representation. In such cases, accurately capturing the system–bath correlations over long timescales necessitates including many effective modes, which would otherwise be computationally prohibitive using full Hilbert space propagation. The use of MPS methods not only reduces the memory requirements by exploiting the area law of entanglement, but also enables efficient time evolution through algorithms like TEBD. As a result, chain mapping combined with tensor network techniques provides a powerful framework for simulating non-Markovian dynamics with controllable numerical convergence.

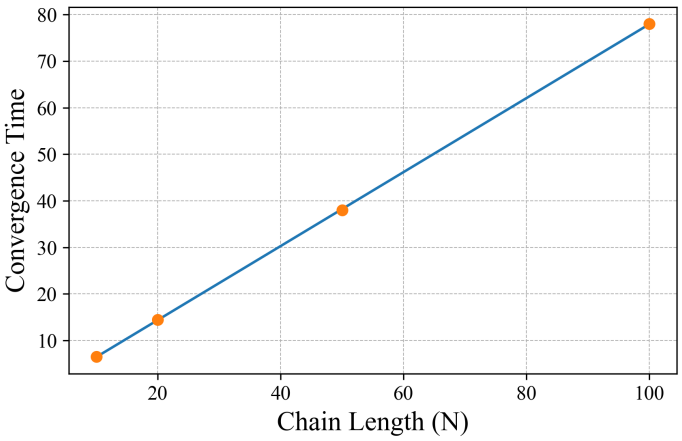



Figure 2.4: Convergence time compared to chain length for all coupling strengths and $\chi_{\max} = 60$. The fitted line (blue) shows a linear increase in convergence time with larger chain lengths, indicating that information and correlations travel at a constant speed down the chain.

### 2.3.2 Convergence with Bond Dimension

The maximum bond dimension $\chi_{\max}$ limits the amount of entanglement that the MPS representation can capture over each chain link. This becomes particularly important in strong coupling regimes where the entanglement between the system and bath grows rapidly and more correlations travel through the system and bath.

To highlight this, we examine the effects of reducing the bond dimension in both weak and strong coupling regimes. Figure 2.5 shows that for all coupling strengths, $\chi_{\max} = 1$ is too small to capture the necessary entanglement between the system and bath. At this minimal bond dimension, the matrix product state representation is essentially a product state, completely neglecting quantum correlations between adjacent sites in the chain, including the system and bath. As a result, the simulation fails to build up required correlations between the system and bath and is unable to accurately reproduce the system's dynamics across all timescales, regardless of coupling strength. Increasing the bond dimension to $\chi_{\max} \geq 2$ leads to a clear improvement in the accuracy of the simulation in the weak coupling regime. This suggests that, for weak system–bath interactions, the entanglement generated between the system and its environment remains relatively low, especially at early times. Consequently, a small bond dimension ($\chi_{\max} \geq 2$) is sufficient to faithfully represent the system dynamics. This behaviour is consistent with expectations that in the weak coupling limit, perturbative approaches such as the Born–Markov approximation are valid precisely because system–environment entanglement is low and decays rapidly.

However, as the coupling strength increases, the required bond dimension grows as well. In the strong coupling regime, entanglement builds up more quickly and persists for longer durations, as the system becomes more strongly correlated with distant bath modes. This requires a higher bond dimension to avoid truncation errors and preserve the non-Markovian features of the dynamics [40]. The scaling of entanglement with time and coupling strength imposes limits on simulations, reinforcing the need for careful convergence checks with respect to bond dimension when using MPS

approaches.

Extending this analysis to the strong coupling regime ($\alpha = 0.1$), we observe a slightly larger dependence on bond dimension. For very short times, $\chi_{\max} = 2$ still agrees with the results from larger bond dimensions, but oscillations quickly emerge as time progresses. This indicates that the strongly coupled spin-boson model exhibits more entanglement than in the weak coupling case. Physically, this is expected as stronger coupling leads to faster and more extensive build-up of correlations between the system and environment, especially as the system starts to develop memory of the dynamics. These system–bath correlations are not localized near the system, but can propagate down the bath into the boundaries of the chain over time, requiring longer-range entanglement to be captured. As a result, the effective entanglement entropy between sections of the chain grows, and a larger bond dimension is required to accurately represent the state using MPS. Failing to account for this growth results in the truncation of physically meaningful degrees of freedom, leading to deviations in the system's evolution, especially at long timescales, where cumulative errors become more significant.

This need for higher bond dimensions is not unique to the spin-boson model. It is a feature of non-Markovian dynamics. As non-Markovianity is described as the emergence of memory effects and backflow of information from the environment to the system, an accurate simulation must allow for enough entanglement. From a computational standpoint, this presents a trade-off: achieving higher accuracy requires increased $\chi_{\max}$, which directly increases memory usage and runtime. While MPS based methods are far more efficient than full state representations, these results emphasize the importance of careful convergence testing not only with respect to chain length and time step, but also bond dimension, especially in regimes where strong system-environment entanglement is expected. This is important to minimise the computation time.

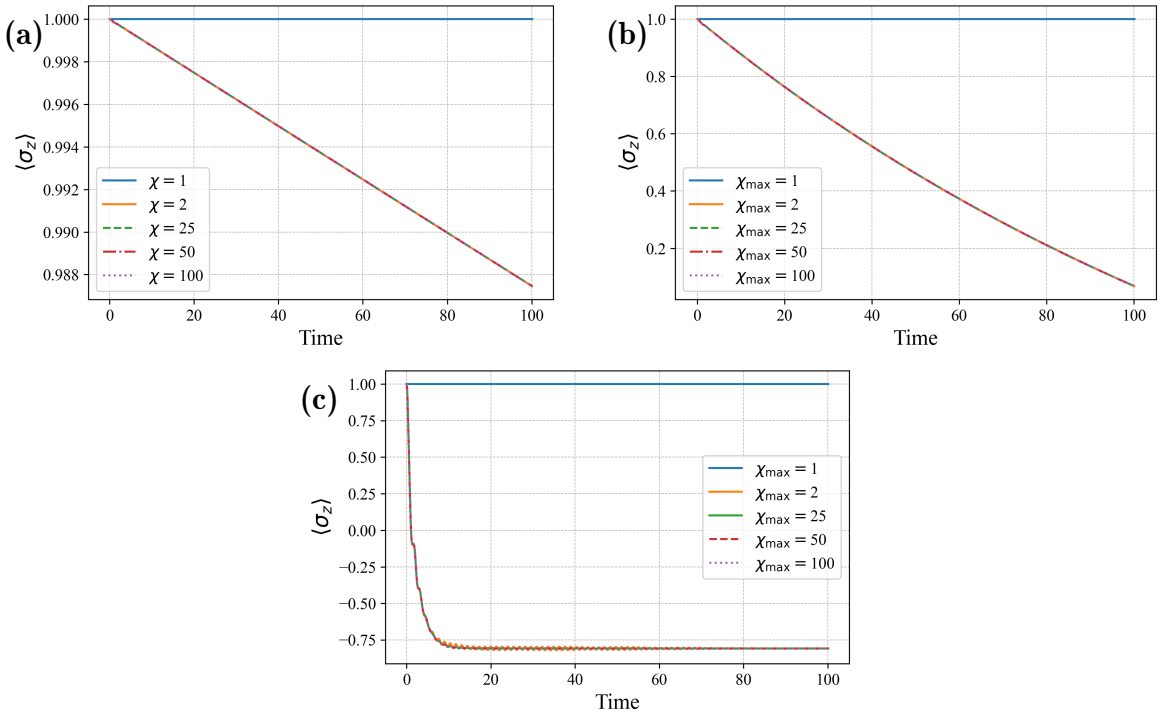



Figure 2.5: Effect of bond dimension on simulation accuracy for different coupling regimes. At $\chi_{\max} = 1$, the dynamics deviate significantly across all timescales, indicating insufficient entanglement capture. Increasing $\chi_{\max}$ improves agreement, especially in the weak coupling cases (a) $\alpha = 10^{-5}$ and (b) $\alpha = 0.001$, where even $\chi_{\max} = 2$ yields convergence. In contrast, stronger coupling (c) $\alpha = 0.1$ requires higher bond dimensions to maintain accuracy over time due to increased entanglement between system and bath. Parameters: $N = 200$

# Chapter 3

# The Surrogate Hamiltonian Method

Simulating open quantum systems (OQS) presents a fundamental challenge due to the intractably large number of environmental degrees of freedom. In principle, accurately modelling a system coupled to its environment requires tracking every single bath mode, an impossible task when dealing with a continuum of modes. However, for sufficiently short timescales, a finite number of bath modes can effectively mimic the influence of the full continuum. This observation underpins the surrogate Hamiltonian approach, first introduced by Baer and Kosloff in 1997 [15], which provides a closed-system approximation to an open system by discretising the bath. The key idea behind this method is both simple and powerful. Rather than attempting to model an infinite environment, a finite set of N bath modes that capture the essential system-bath interactions are selected. Over short enough timescales, this discrete set of modes behaves as a quasi-continuum, effectively reproducing the bath's influence on the system dynamics. What makes this approach particularly valuable is that, unlike many approximation schemes that simply eliminate the bath degrees of freedom, the surrogate Hamiltonian method preserves the dynamical evolution of both system and bath. This preservation of bath dynamics makes the method especially well-suited for investigating non-Markovian effects, where memory effects and information backflow from the environment play crucial roles [4].

This chapter introduces the surrogate Hamiltonian method and develops the theoretical framework necessary for the remainder of this thesis. The chapter is structured as follows: Firstly, the fundamental concepts behind the surrogate Hamiltonian and details on how such a Hamiltonian can be systematically constructed are outlined. This section draws heavily from Ref. [41], which provides many of the technical details. The next section then applies these concepts to the same spin-boson model shown in chapter 2 constructing a surrogate Hamiltonian for this model. The final section presents the convergence behaviour of the model and discusses the limitations of the surrogate Hamiltonian.

## 3.1 The concept of the Surrogate Hamiltonian

At its core, the Surrogate Hamiltonian approach truncates the infinite bath of an OQS to model only its $N$ most strongly interacting modes, propagating the system and surrogate bath together as a closed system. This transforms the original Hamiltonian into a finite-dimensional approximation:

$$\hat{H} = \hat{H}_{\mathrm{S}} + \hat{H}_{\mathrm{B}} + \hat{H}_{\mathrm{I}} \rightarrow \hat{H}^{\mathrm{Sur}} = \hat{H}_{\mathrm{S}} + \hat{H}_{\mathrm{B}}^{\mathrm{Sur}} + \hat{H}_{\mathrm{I}}^{\mathrm{Sur}} \tag{3.1}$$

where each component must be reconstructed for the reduced Hilbert space. While the system Hamiltonian $\hat{H}_S$ acts only on the system degrees of freedom and remains formally unchanged, it must be embedded in the smaller total Hilbert space as $\hat{H}_S \otimes \mathbb{I}_B^{Sur}$ where $\mathbb{I}_B^{Sur}$ is the identity operator for the $N$-mode surrogate bath. The system dynamics are then recovered by tracing out the surrogate bath modes post-propagation. This approach leverages the fact that the bath's importance lies not in the dynamics of individual modes but in their collective influence on the system evolution, allowing considerable flexibility in how the bath can be modelled. To construct the surrogate bath, the $N$ most strongly interacting modes are chosen, transforming the bath Hamiltonian to a finite sum

$$\hat{H}_{\mathrm{B}} = \sum_{n=1}^{\infty} \omega_n \hat{a}_n^\dagger \hat{a}_n \rightarrow \sum_{n=1}^{N} \omega_n^{Sur} \hat{a}_n^\dagger \hat{a}_n = \hat{H}_{\mathrm{B}}^{\mathrm{Sur}}, \tag{3.2}$$

with $\omega_n^{\mathrm{Sur}}$ being the frequencies of the sampled modes and $\hat{a}_n^\dagger$ and $\hat{a}_n$ being the creation and annihilation operators acting on the nth bath mode. In the limit $N \rightarrow \infty$, the surrogate Hamiltonian recovers the full infinite bath, ensuring that the approximation becomes exact with infinite modes. Next the interaction component of the Hamiltonian $\hat{H}_I^{\mathrm{Sur}}$ must be constructed. For infinitely many modes, the interaction Hamiltonian can be described

$$\hat{H}_I = \hat{A}_S \int_0^{\omega_c} d\omega \; h(\omega)(\hat{a}(\omega) + \hat{a}^\dagger(\omega)), \tag{3.3}$$

where $\hat{A}_S$ is some coupling operator acting on the system and $h(\omega)$ is the coupling function for the bath. In the continuous limit, the coupling function is defined

$$J(\omega) = \pi(h(\omega))^2 \rho(\omega) \rightarrow h(\omega) = \sqrt{\frac{2J(\omega)}{\pi\rho(\omega)}}, \tag{3.4}$$

where $J(\omega)$ is the spectral density. In this thesis, a factor of $\frac{2}{\pi}$ is added to the spectral density to ensure continuity between the chain mapping and surrogate Hamiltonian methods. This factor arises from the Fourier transformation and may or may not be needed depending on the spectral density definition used. In the continuous limit, the spectral density $J(\omega)$ characterizes the density of bath modes and their coupling strength at each frequency, when discretizing the bath into $N$ modes with frequencies $\{\omega_n^{\mathrm{Sur}}\}$, each discrete mode must represent a finite frequency interval $\Delta\omega_n = \omega_{n+1} - \omega_n$. To preserve the total coupling strength, the contribution from each discrete mode should equal the integrated contribution from the corresponding continuous frequency interval [3]. This requirement gives the condition

$$h^{\mathrm{Sur}}(\omega_n^{\mathrm{Sur}}) = \sqrt{\frac{2}{\pi} J(\omega_n^{\mathrm{Sur}}) \Delta\omega_n^{\mathrm{Sur}}}, \tag{3.5}$$

where $\Delta\omega_n^{\mathrm{Sur}}$ is the gap between the sampled frequencies. This gap is inversely proportional to the density of states for a finite bath $\rho(\omega_n^{\mathrm{Sur}})$

$$\rho(\omega_n^{\mathrm{Sur}}) = \frac{1}{\Delta\omega_n^{\mathrm{Sur}}} = \frac{1}{\omega_{n+1}^{\mathrm{Sur}} - \omega_n^{\mathrm{Sur}}}. \tag{3.6}$$

Therefore, the coupling function can be rewritten,

$$h^{\mathrm{Sur}}(\omega_n^{\mathrm{Sur}}) = \sqrt{\frac{2}{\pi} J(\omega_n^{\mathrm{Sur}})/\rho(\omega_n^{\mathrm{Sur}})}, \tag{3.7}$$

resulting in the discretised interaction Hamiltonian

$$\hat{H}_I = \hat{A}_s \sum_{n=1}^{N} h^{\mathrm{Sur}}(\omega_n^{\mathrm{Sur}})(\hat{a}_n + \hat{a}_n^\dagger). \tag{3.8}$$

This discretisation ensures that when the number of modes increases and the frequency spacing decreases, the total system-bath coupling strength is constant. This results in the interaction Hamiltonian converging to the continuous definition in the infinite limit.

Now with the surrogate Hamiltonian constructed, the most strongly interacting modes must be selected. Careful consideration must be taken when choosing which mode frequencies are sampled in the surrogate bath $\omega_n^{Sur}$. The frequency range must be wide enough to include the most strongly interacting modes but not so wide that large gaps appear between mode frequencies. This finite sampling works under the assumption that modes close to resonance dominate the system's short-time behaviour and that when sampling close to resonance, the bath retains sufficient resolution near the frequencies where the coupling strength is highest [42]. There are many methods to sample frequencies for the surrogate bath, one simple approach is to sample frequencies equidistantly and centred around the system frequency $\omega_0$ with one mode positioned exactly on resonance. The other modes are then distributed equidistantly either side of the system frequency. The spacing between modes is determined by:

$$\Delta\omega_N^{\mathrm{Sur}} = \frac{\omega_c - \omega_0}{\lfloor N/2 \rfloor}, \tag{3.9}$$

where $\lfloor N/2 \rfloor$ represents the number of modes on each side of the central resonant mode. The individual mode frequencies are then given by:

$$\omega_n^{\mathrm{Sur}} = \omega_c - n\Delta\omega_N^{\mathrm{Sur}}. \tag{3.10}$$

In this sampling approach, the sampling range spans $[\omega_0 - x, \omega_0 + x]$, where $x = \omega_c - \omega_0$. This parameter must be chosen carefully to ensure that the most strongly interacting modes are included while maintaining adequate frequency resolution. Since bath frequencies must remain positive, the lower bound cannot extend below zero, this also constrains the cut-off frequency. The maximum cut-off frequency occurs in the case where the range extends down to zero frequency, this results in a sampling window of $[0, 2\omega_0]$, imposing a maximum cut-off frequency of $\omega_c = 2\omega_0$.

To maintain consistency across different approximations, the sampling range must remain fixed while the number of modes $N$ varies. As more modes are added, the density of sampled frequencies increases proportionally, providing better resolution of the most important frequency region around $\omega_0$. This ensures that comparisons between different bath sizes approximate the same physical frequency range with varying levels of density. Although simple, this method offers a reasonable approximation when the spectral density varies smoothly and dynamics are dominated by a narrow frequency window around $\omega_0$.

### 3.1.1 Self Convergence

Much like the chain mapping method, one main limitation of the surrogate Hamiltonian approximation is the emergence of unphysical recurrences caused by the finite environment.

Self-convergence in a surrogate Hamiltonian is achieved when increasing the bath size $N$ no longer changes the short-time dynamics of the approximation. The convergence time for a given surrogate Hamiltonian refers to how long the approximated dynamics remain accurate before finite-size effects cause deviation from the true evolution. This self-convergence indicates that the most important bath modes have been captured for the relevant timescales. However, it is important to note that larger bath sizes only extend the duration of accurate dynamics, they do not prevent the inevitable divergence that occurs when boundary reflections reach the system. This relationship between bath size and convergence time allows one to optimize the number of modes based on the desired simulation duration.

Extensions of the surrogate Hamiltonian method have been proposed to delay or remove these boundary effects, in particular the stochastic surrogate hamiltonian which swaps bath modes with an secondary thermal reservoir to reduce the amount of recurrences [43].

### 3.1.2 Reducing Computational Complexity: Limiting Excitations

The Hilbert space of a surrogate Hamiltonian with $N$ modes scales as $2^N$, which quickly becomes computationally intractable. One can keep the Hilbert space manageable by limiting $N$, but this also limits the time scale your approximation is convergent for. However there is another approach to reduce the Hilbert space. Since the likelihood of every mode in the bath being excited at the same time is low, one can restrict the number of simultaneous excitations allowed in the bath and only propagate a hamiltonian with this reduced excitation subspace. Depending on how many simultaneous excitations are allowed controls how big the Hilbert space is. For example, consider a surrogate bath of $N = 3$ two-level systems. Without any restrictions, the bath alone has $2^3 = 8$ possible configurations:

$$\downarrow\downarrow\downarrow, \quad \downarrow\downarrow\uparrow, \quad \downarrow\uparrow\downarrow, \quad \downarrow\uparrow\uparrow, \quad \uparrow\downarrow\downarrow, \quad \uparrow\downarrow\uparrow, \quad \uparrow\uparrow\downarrow, \quad \uparrow\uparrow\uparrow\,.$$

However, especially in weak coupling ($\alpha \ll 1$) and at low temperatures states with multiple excitations are unlikely. The probability of creating multiple excitations in the bath is suppressed by factors of $\alpha^n$ for $n$ excitations, making multi-excitation states negligible. Restricting the bath to at most one excitation, we reduce the number of possible states from 8 states to just 4

$$\downarrow\downarrow\downarrow, \quad \downarrow\downarrow\uparrow, \quad \downarrow\uparrow\downarrow, \quad \uparrow\downarrow\downarrow\,.$$

Limiting to a single excitation space transforms the bath's Hilbert space from exponential scaling ($2^N$) to linear scaling ($N+1$). For the total system-bath Hilbert space, this means going from $2 \times 2^N$ dimensions to just $2 \times (N + 1)$ dimensions. The linear scaling is particularly advantageous when a large surrogate bath is required. For example, with $N = 100$ modes the full excitation space has a Hilbert space $\sim 10^{30}$

states, this is impossible to propagate. However, reducing possible states to a single excitation subspace reduces the Hilbert space to $\sim 100$ states, which is easily propagated. Of course, this approximation has its limits. For stronger coupling or longer evolution times, processes that create multiple bath excitations become important. In these cases you can continue using a limited excitation subspace but increase the number of allowed excitations. In these cases, the Hilbert space scales according to the binomial coefficient, e.g. for the three excitation subspace the Hilbert space scales $\sim \binom{N}{3}$.

## 3.2 Approximating the Spin-Boson Model

To demonstrate the Surrogate Hamiltonian method, we apply it to the spin-boson model, the same system analyzed using Chain Mapping in Chapter 2. This parallel treatment allows for direct comparison between the two approaches and provides a benchmark against established analytical and numerical results. The specific numerical implementation details, including basis construction and propagation algorithms, are provided in Appendix A.

We begin with the spin-boson Hamiltonian given in Eq. (2.7), where a two-level system with frequency $\omega_0$ interacts with a bosonic environment characterized by the hard cut-off spectral density in Eq. (2.8). The key parameters for our analysis are the system frequency $\omega_0 = 1$, coupling strength $\alpha$, and cutoff frequency $\omega_c$. Following the truncation procedure outlined in the previous section, we select $N$ bath modes with frequencies $\omega_n^{\text{Sur}}$ sampled equidistantly around the system frequency. The frequency range spans $[\omega_0 - x, \omega_0 + x]$ where $x = \omega_0$ to ensure positive frequencies, giving a maximum cutoff of $\omega_c = 2\omega_0$. This sampling strategy is based on the assumption that modes close to resonance provide the dominant contribution to the system dynamics in the weak to moderate coupling regime.
The resulting surrogate Hamiltonian takes the form:

$$\hat{H}^{\text{Sur}} = \omega_0 \sigma_z + \sum_{n=1}^{N} \omega_n^{\text{Sur}} \hat{a}_n^\dagger \hat{a}_n + \frac{1}{2}\sigma_x \sum_{n=1}^{N} h^{\text{Sur}}(\omega_n^{\text{Sur}})(\hat{a}_n + \hat{a}_n^\dagger), \tag{3.11}$$

where the first term represents the free system evolution, the second term describes the truncated bath dynamics, and the third term captures the system-bath interaction. The coupling function is constructed to preserve the spectral properties of the original continuous bath:

$$h^{\text{Sur}}(\omega_n^{\text{Sur}}) = \sqrt{4\alpha\omega_n^{\text{Sur}}/\rho(\omega_n^{\text{Sur}})}, \tag{3.12}$$

where $\rho(\omega_n^{\text{Sur}}) = 1/\Delta\omega$ is the density of states with $\Delta\omega$ being the constant frequency spacing between adjacent modes. The factor of $4\alpha$ rather than $2\pi\alpha$ arises from the specific frequency normalization used in the discretization scheme, ensuring that the discrete spectral density approaches the continuous limit as $N \to \infty$.
With the surrogate Hamiltonian constructed, we have a finite-dimensional closed system that can be propagated by solving the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}^{\text{Sur}}|\psi(t)\rangle. \tag{3.13}$$

We initialize the system in the excited spin state with all bath modes in their ground state:

$$|\psi_{\text{Sur}}(0)\rangle = |\uparrow\rangle \otimes |0_1, 0_2, \ldots, 0_N\rangle, \tag{3.14}$$

corresponding to a factorized initial condition with no initial system-bath entanglement.
The time evolution is performed using QuTiP's `sesolve` function. System observables such as $\langle\sigma_z(t)\rangle$ are computed by tracing out the bath degrees of freedom:

$$\langle\hat{O}_S(t)\rangle = \mathrm{Tr}_B[\hat{O}_S \otimes \mathbb{I}_B \cdot \rho_{\mathrm{Sur}}(t)], \tag{3.15}$$

where $\rho_{\mathrm{Sur}}(t) = |\psi_{\mathrm{Sur}}(t)\rangle\langle\psi_{\mathrm{Sur}}(t)|$. The accuracy of the approximation is assessed through self-convergence (independence of results from bath size $N$ for times shorter than the recurrence time) and comparison to Markov approximation results in the weak coupling regime. In the following section, we present the initial results of these simulations, focusing on the self convergence properties and limitations arising from finite bath size effects.

## 3.3 Convergence Analysis

As in the Chain Mapping approach, the convergence time in the Surrogate Hamiltonian method is governed by the number of bath modes, $N$. Increasing the size of the surrogate bath delays the onset of recurrences due to finite-size effects, thereby extending the timescale over which the simulation remains accurate. However, achieving convergence is further complicated by the strict criteria that the sampled modes must satisfy to ensure internal consistency and self-convergence of the method. In this analysis, we only use odd bath sizes as including a resonant mode is crucial to maximising convergence time.

### 3.3.1 Convergence with Number of Modes

Using the spin-boson model described above, we begin with a spin in its up state $|\uparrow\rangle$ coupled to a bath with all modes in the ground state $|0000\ldots\rangle$. For all the simulations in this section $\omega_0 = 1$, $\omega_c = 2$, and $T = 0$.

The fundamental limitation of the Surrogate Hamiltonian approach becomes apparent when examining how the approximation convergence time depends on the number of bath modes included. Figure 3.1 demonstrates the characteristic behaviour of finite bath truncation by showing the time evolution of $\langle\sigma_z(t)\rangle$ for different bath sizes $N$ in the weak coupling regime ($\alpha = 0.001$). Initially, all curves follow nearly identical trajectories, indicating that the short-time dynamics are well-captured even by relatively small bath sizes. This initial agreement reflects the fact that the most strongly coupled modes near resonance dominate the early-time evolution, validating the assumption underlying the frequency sampling strategy.

However, as time progresses, a clear hierarchy emerges where smaller bath sizes begin to deviate from the dynamics predicted by larger systems. This divergence occurs at a time that increases with bath size, revealing the finite-size effects inherent to the truncation. The physical origin of this breakdown lies in the fact that correlations and energy initially released into the bath modes eventually reach the artificially imposed boundaries of the finite surrogate bath. In an infinite bath, these correlations would continue propagating away from the system indefinitely, but in the truncated approximation, they reflect back toward the system, causing unphysical recurrences

that corrupt the approximated dynamics.

The coupling strength dependence of this convergence behaviour is illustrated in figure 3.2, which compares weak coupling ($\alpha = 10^{-5}$) and strong coupling ($\alpha = 0.1$) regimes. The convergence times show only small variation between these very different coupling regimes, suggesting that the finite-size effects are primarily determined by the bath structure and sampling rather than the strength of the system-bath interaction. In the weak coupling case (a), the decay is clearly visible and the finite-size recurrences appear as clear discontinuities in the otherwise smooth evolution. For strong coupling (b), the dynamics become more complex as a larger amount of correlations and information are sent into the bath, causing a larger amount of artificial recurrences, but the fundamental pattern of initial convergence followed by finite-size breakdown remains similar.

This weak dependence on coupling strength is quantified more systematically in Figure 3.3, which plots the convergence time as a function of coupling strength $\alpha$. The convergence time shows a gradual decrease with increasing coupling strength, but the effect is relatively modest compared to the dramatic changes in the actual system dynamics. This behaviour can be understood by recognizing that stronger coupling primarily affects the amplitude and character of the correlations propagating into the bath, rather than their propagation speed, which is determined by the bath's intrinsic frequency scales. The computational implications of finite bath size are significant for practical applications. While increasing $N$ extends the valid simulation time, it also exponentially increases the Hilbert space dimension. This scaling severely limits the achievable bath sizes, particularly for longer simulation times. However, limiting the excitation subspace will increase achievable bath sizes.

One important limitation revealed by these convergence studies is that the surrogate Hamiltonian approach cannot easily distinguish between genuine non-Markovian effects and artifacts arising from finite bath size. Both phenomena can produce similar signatures such as non-exponential decay and apparent memory effects. This ambiguity requires careful analysis of the convergence properties and comparison with other methods to ensure that observed non-Markovian behaviour reflects true physics rather than numerical artifacts. The method works when the genuine physical timescales of interest are shorter than the convergence time set by the finite bath boundaries.

### 3.3.2 Effects of Sampling Method on Self Convergence

The choice of frequency sampling strategy proves crucial for the accuracy and efficiency of the Surrogate Hamiltonian approximation.

Figure 3.4 demonstrates the consequences of sampling frequencies that are not centred around the system frequency $\omega_0$. In this figure, the bath modes are sampled equidistantly between 0 and $\omega_c$ with an additional resonant mode at $\omega_0$ included. Despite the presence of the resonant mode, this sampling strategy fails to achieve self-convergence as the bath size increases, unlike the centred and equidistant sampling approach discussed previously. The lack of self-convergence means that increasing the number of bath modes continues to change the short-time dynamics, indicating that the approx-

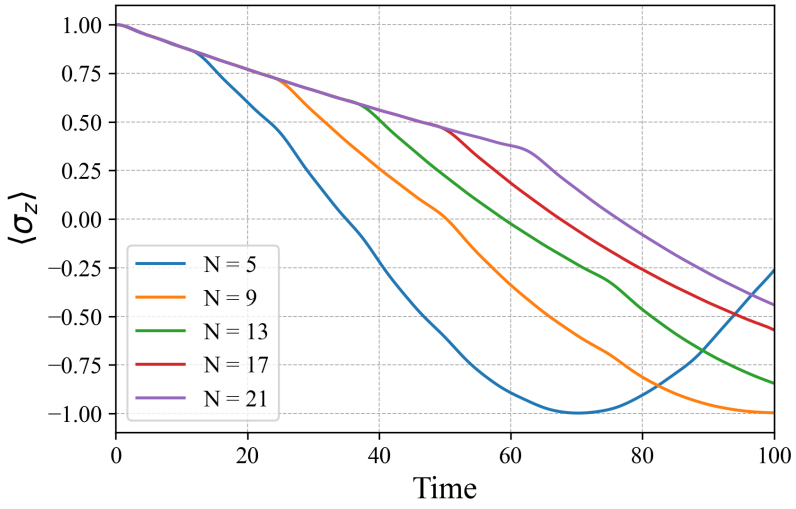



Figure 3.1: Surrogate Hamiltonian convergence with bath size: Time evolution of $\langle \sigma_z(t) \rangle$ for different numbers of bath modes $N$, showing initial agreement followed by divergence at characteristic recurrence times. Smaller bath sizes deviate earlier due to finite-size boundary effects. Parameters: $\alpha = 0.001$, $\omega_0 = 1$, $\omega_c = 2$.

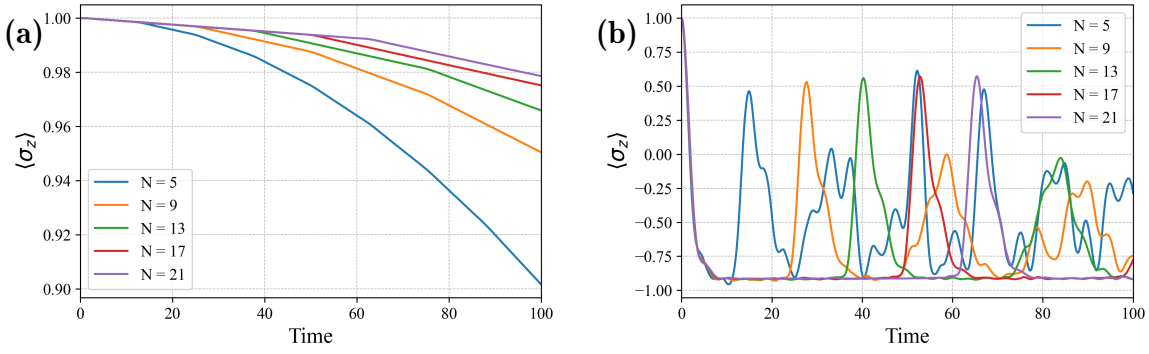



Figure 3.2: Bath size convergence across coupling regimes: Time evolution of $\langle \sigma_z(t) \rangle$ for different bath sizes $N$ in (a) weak coupling ($\alpha = 10^{-5}$) and (b) strong coupling ($\alpha = 0.1$) regimes. Both regimes show similar convergence patterns with slightly shorter recurrence times for stronger coupling. The oscillatory behaviour in (b) reflects additional amount of recurrences that emerge in the strong coupling limit.

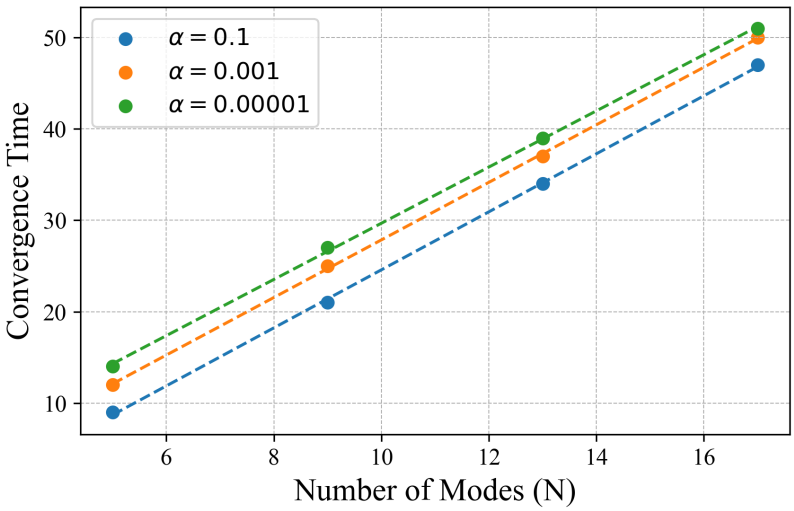



Figure 3.3: Coupling strength dependence of convergence time: The time at which finite-size effects cause deviation from converged dynamics shows a weak dependence on coupling strength $\alpha$, decreasing gradually with stronger coupling. This suggests that recurrence times are primarily determined by bath structure rather than coupling strength. Data taken from figures 3.1 and 3.2.Parameters: $\omega_0 = 1$, $\omega_c = 2$.

imation has not captured the essential physics even for early times. This behaviour prevents reliable extrapolation to the infinite bath limit and undermines confidence in the results.

The fundamental issue lies in how the sampling density changes as more modes are added. In the non-centred approach, adding modes equidistantly from 0 to $\omega_c$ and then later adding a resonant mode creates an uneven distribution and thus uneven density of states. While including the resonant mode is crucial for maximising convergence time, it disrupts the uniform spacing and creates an asymmetric density distribution around $\omega_0$. As the total number of modes increases, the density of states near the resonant frequency does not scale proportionally with the overall bath size. This uneven scaling means that the spectral weight in the most important frequency region around $\omega_0$ is not adequately resolved, preventing the approximation from converging to the correct infinite-bath limit regardless of how many modes are included.

In contrast, the centred sampling strategy maintains a constant frequency range while increasing the density uniformly across the important frequency window around $\omega_0$. This ensures that as $N$ increases, the resolution improves proportionally in the spectral region that matters most for the system dynamics. The sampling density must scale appropriately with the number of modes to preserve the correct spectral weight distribution and achieve convergence.

This sensitivity to sampling strategy highlights a fundamental constraint of the surrogate Hamiltonian approach. Unlike methods such as chain mapping that systematically capture the most important correlations through their mathematical structure, the direct frequency sampling approach requires prior knowledge of which frequency

ranges dominate the physics. Moreover, it demands careful attention to how the sampling density scales with bath size to ensure that the approximation self converges to the correct infinite-bath limit.

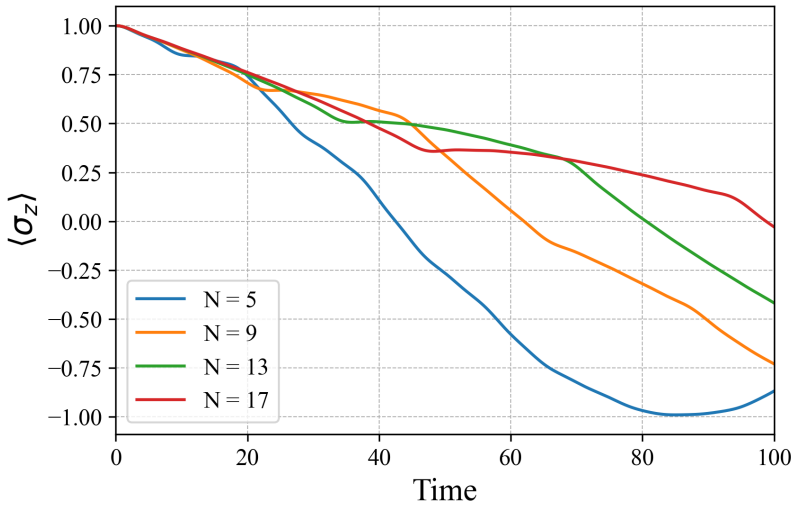



Figure 3.4: Failure of non-centred frequency sampling: Time evolution of $\langle \sigma_z(t) \rangle$ when bath modes are sampled equidistantly from 0 to $\omega_c$ rather than centred around $\omega_0$. Despite including a resonant mode, this sampling strategy fails to achieve self-convergence, with dynamics continuing to change as bath size increases. This is likely because not all modes are equidistant, the added resonant mode changes the distribution. Parameters: $\alpha = 0.001$, $\omega_0 = 1$, $\omega_c = 5$.

# Chapter 4

# Comparing the methods

This chapter presents a systematic comparison of the Surrogate Hamiltonian and Chain Mapping methods introduced in the previous chapters. Their performance is analysed against the Markovian approximation across different coupling regimes. This analysis focuses on the spin-boson model discussed in chapters 2 and 3, allowing direct comparison between methods using the same physical system. Using bath sizes that are self converged for the timescales analysed, the surrogate Hamiltonian and chain mapping methods are analysed across weak and strong coupling regimes.

For the Markov approximation benchmark, we construct a master equation for the same spin-boson model with the equations for decay rate from Eq. 1.5 and the corresponding jump operator from Eq. 1.6. This is implemented using QuTiP's `mesolve` function. The Markov approximation provides a reference point for comparison, particularly in the weak coupling regime where the description remains valid.

Throughout this chapter, we track the system dynamics through the expectation value of the Pauli-$z$ operator, $\langle\sigma_z\rangle$. This choice of observable is particularly convenient for our spin-boson model. Since the system Hamiltonian commutes with $\sigma_z$ and we initialize the system in an eigenstate of $\sigma_z$, the dynamics remain confined to the $z$-axis of the Bloch sphere, with $\langle\sigma_x\rangle = \langle\sigma_y\rangle = 0$ for all times. Furthermore, in this restricted dynamics, $\langle\sigma_z\rangle$ provides a direct measure of non-Markovianity through its connection to the state space volume [23]. Any increase in $\langle\sigma_z\rangle$ corresponds to an expansion of the accessible state space volume. This is a definitive signature of information backflow from the environment to the system which cannot occur in Markovian dynamics where the state space volume must decrease monotonically.

The chapter is organized as follows: the weak coupling regime is first examined, where a reliable benchmark is provided by the Markov approximation for validating the accuracy of both approximation methods. Through this analysis, the effects of cut-off frequency selection on method performance are revealed, and the fundamental constraints imposed by different frequency sampling strategies are highlighted. The strong coupling regime is then investigated. In this regime non-Markovian dynamics begin to take effect and the assumptions made in the Markov approximation are no longer valid. The differences between each approximation method is then analysed and compared to what we would expect at strong coupling.

## 4.1 Weak Coupling Regime

In the weak coupling regime ($\alpha \leq 0.01$), the system-bath interaction is sufficiently small that the Markov approximation provides a reliable description of system dynamics. We use this regime as a test of method approximation accuracy, as methods designed to capture non-Markovian dynamics must first reproduce Markovian behaviour when memory effects are negligible (the Markovian limit). We therefore begin our comparative analysis in this limit, where analytical solutions are available and the physics is well understood.

The fundamental challenge in modelling open quantum systems lies in representing an infinite-dimensional bath with finite computational resources. Each method addresses this challenge differently, leading to distinct sensitivities and limitations. Figure 4.1 illustrates how the Surrogate Hamiltonian and Chain Mapping approaches construct their finite bath representations through contrasting frequency sampling strategies.

### 4.1.1 Sampling Strategies

The Surrogate Hamiltonian method adopts a physically intuitive approach by sampling bath mode frequencies symmetrically around the system frequency, $\omega_0 = 1$, at equal intervals. This strategy relies on the assumption that modes near resonance contribute most significantly to the system–bath interaction, an assumption valid in many physical contexts. However, the requirement of symmetric sampling imposes a geometric constraint: to avoid unphysical negative frequencies, the cut-off frequency must satisfy $\omega_c \leq 2\omega_0$. In our case, this limits the cut-off to $\omega_c \leq 2$, which becomes restrictive when attempting to model environments with broad spectral densities. The noticeable impact of $\omega_c$ in this system likely arises from the use of a hard cut-off spectral function, in which mode populations are uniform across the range $[0, \omega_c]$. In contrast, a soft or exponential cut-off where high-frequency contributions decay more gradually, and are likely less affected by insufficient sampling at high frequencies.
In contrast, the Chain Mapping method adopts a more mathematically based approach based on the structure of the transformation operator. Rather than choosing frequencies based on resonance proximity, this method uses orthogonal polynomial recursion relations to construct a non-uniform discretisation of the bath. This is used to ensure the chain modes represent the most dominant correlations in the original OQS. For a hard cut-off spectral density, this procedure naturally concentrates bath modes near $\omega_c/2$ through its use of Jacobi polynomials. Importantly, this mathematically derived sampling adjusts automatically with the chosen cut-off frequency, maintaining consistent resolution and accuracy across a range of spectral parameters without requiring manual tuning.

### 4.1.2 Weak Coupling Dynamics

These structural differences manifest clearly in the dynamics shown in Figure 4.3. The Markov approximation, serving as our reference, exhibits negligible dependence on $\omega_c$, with identical dynamics for $\omega_c = 2, 5$, and 7 at the resolution shown. This invariance is expected. As the Markov approximation incorporates the cut-off only through the

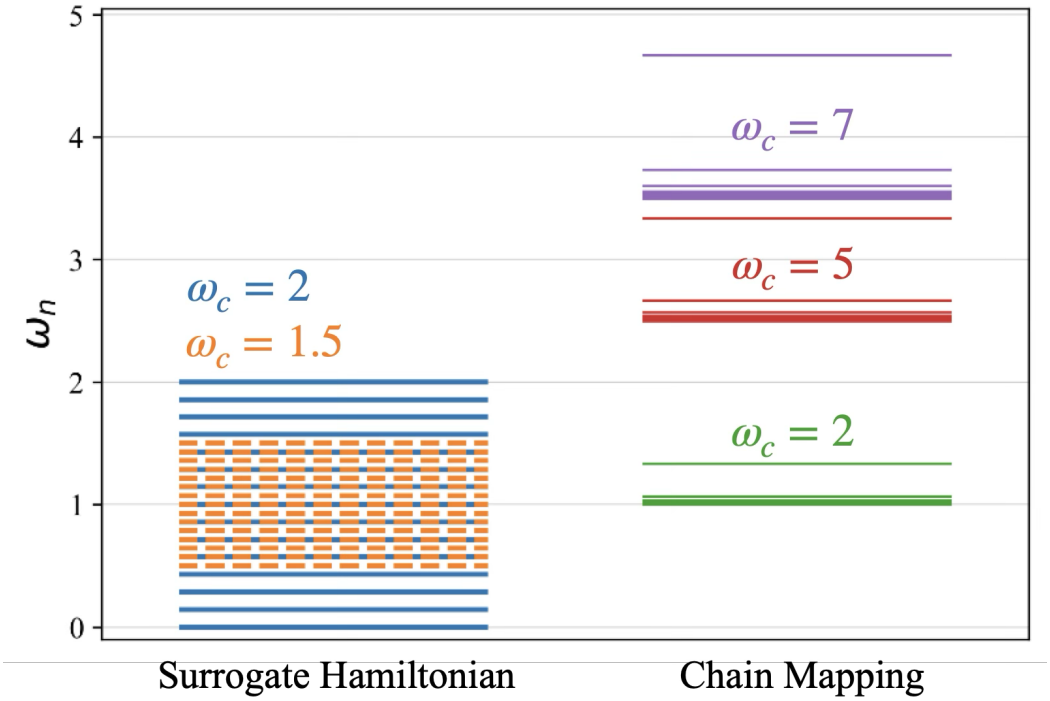



Figure 4.1: Comparison of frequency sampling strategies between Surrogate Hamiltonian and Chain Mapping methods for different cut-off frequencies $\omega_c$. Left side shows the Surrogate Hamiltonian approach, which samples frequencies equidistantly around the system frequency $\omega_0 = 1$. This is compared to the Chain Mapping approach on the right, where chain frequencies are determined using recursion coefficients from the Jacobi polynomials used to map the system onto a one-dimensional chain [24]. Rather than sampling modes that are physically close in frequency to the system, the chain mapping approach selects mode frequencies such that the most important correlations are captured in the chain. This contrast highlights the fundamental difference in how each method selects and distributes the important bath modes. Parameters: Surrogate Hamiltonian $N = 15$, Chain Mapping $N = 50$.

spectral density integral, not through explicit mode selection. In contrast, both effective mode methods show strong frequency dependence, reflecting their reliance on finite bath representations.

The Chain Mapping method demonstrates robustness across the entire range of cut-off frequencies tested. For $\omega_c \geq 5$, it achieves excellent agreement with the Markovian dynamics throughout the evolution. Table 4.1 quantifies this agreement, showing that Chain Mapping yields decay rates identical to the Markov approximation in the short time regime ($t = 10$–$20$) for all cut-off values. Minor deviations appear only for $\omega_c = 2$, and these are confined to the very short time dynamics.

The Surrogate Hamiltonian tells a different story. At its maximum allowed cut-off of $\omega_c = 2$, it produces dynamics qualitatively similar to Chain Mapping, yet both methods deviate slightly from the Markov result. However, the decay rate is quite similar to both the Markov and chain mapping approach. This mutual agreement while missing the expected dynamics indicates that $\omega_c = 2$ may provide insufficient spectral coverage for either approach to capture the full dynamics, but it still approximates quite well. The situation deteriorates for $\omega_c = 1.5$, where the Surrogate Hamiltonian's compressed sampling range $[0.5, 1.5]$ fails to capture essential bath correlations. The decay rate drops to nearly half the Markovian value (Table 4.1), indicating a slowing in dissipation. This prediction of incorrect dynamics persists even at extremely weak coupling ($\alpha = 10^{-5}$, Figure 4.3a).

The physical origin of the Surrogate Hamiltonian's failure becomes clear when we consider the spectral density $J(\omega) = 2\pi\alpha\omega_c^{1-s}\omega^s$. For ohmic dissipation ($s = 1$), the density increases linearly with frequency up to the cutoff. When $\omega_c = 1.5$, the narrow sampling window $[0.5, 1.5]$ captures only a small fraction of the total spectral weight, missing the low frequency modes. The Chain Mapping method avoids this pitfall through its adaptive frequency distribution, which automatically adjusts to capture the relevant spectral features regardless of the cut-off value.

A subtle but important feature appears in the vert short-time dynamics (Figure 4.3, inset). Both effective mode methods exhibit an initial slippage, a brief period where $\langle\sigma_z\rangle$ decreases more slowly than predicted by the Markov approximation. This phenomenon occurs because the system and bath begin in an uncorrelated product state, and finite time is required to establish the correlations that drive dissipation [44]. Interestingly, Chain Mapping with larger cut-offs ($\omega_c \geq 5$) displays a brief acceleration following the initial slowdown, partially compensating for the slippage effect. This behaviour suggests that the polynomial-based frequency selection captures additional short-time correlations when $\omega_c$ is larger. While the standard Markov approximation neglects these initial correlation effects entirely, extensions exist that incorporate them [45]. For our purposes, these short-time deviations do not compromise either method's validity, as both converge to the correct intermediate-time behaviour where most physical observables of interest are measured.

The practical implications of these differences are significant. When the system frequency $\omega_0$ is small, the geometric constraints of the Surrogate Hamiltonian approach, specifically the $\omega_c \leq 2\omega_0$ limitation, severely restrict the range of valid spectral cutoffs,

thereby limiting its applicability in broad-band environments. In contrast, the Chain Mapping method, being free from such geometric constraints, consistently delivers accurate results across a wider range of parameter regimes. However, this generality comes at the cost of computational efficiency. As illustrated in Figures 2.4 and 3.3, the Chain Mapping approach requires a larger number of modes (e.g., $N = 50$) to achieve convergence compared to the Surrogate Hamiltonian, which achieves similar temporal coverage with just $N = 15$ modes. This suggests that the Surrogate Hamiltonian's direct, resonance focused sampling is more efficient in capturing short-to-intermediate time dynamics. Nevertheless, this advantage is offset by its exponential scaling in the full excitation subspace, making large surrogate baths computationally intractable, especially at sizes comparable to those routinely used in the Chain Mapping approach.

These structural differences manifest clearly in the methods' performance. The Chain Mapping approach shows remarkable stability across different cut-off values because its polynomial-based selection inherently captures the most important correlations regardless of $\omega_c$. The Surrogate Hamiltonian, constrained by its geometric limitations and uniform sampling, must compromise between spectral coverage and sampling density. For $\omega_c = 2$, both methods yield similar dynamics, suggesting this represents the minimum bath size needed for accurate short-time evolution. However, the Surrogate Hamiltonian cannot benefit from larger cut-off values due to its sampling constraints, while Chain Mapping continues to improve with increasing $\omega_c$.

This analysis underscores a key trade-off: the Surrogate Hamiltonian's physically motivated sampling provides intuitive parameter selection and simple construction but imposes rigid constraints, while Chain Mapping's mathematical framework offers greater flexibility at the cost of less transparent mode selection and more complicated application. For systems where $\omega_0$ is small relative to the bath's spectral width, these constraints severely limit the Surrogate Hamiltonian's applicability.

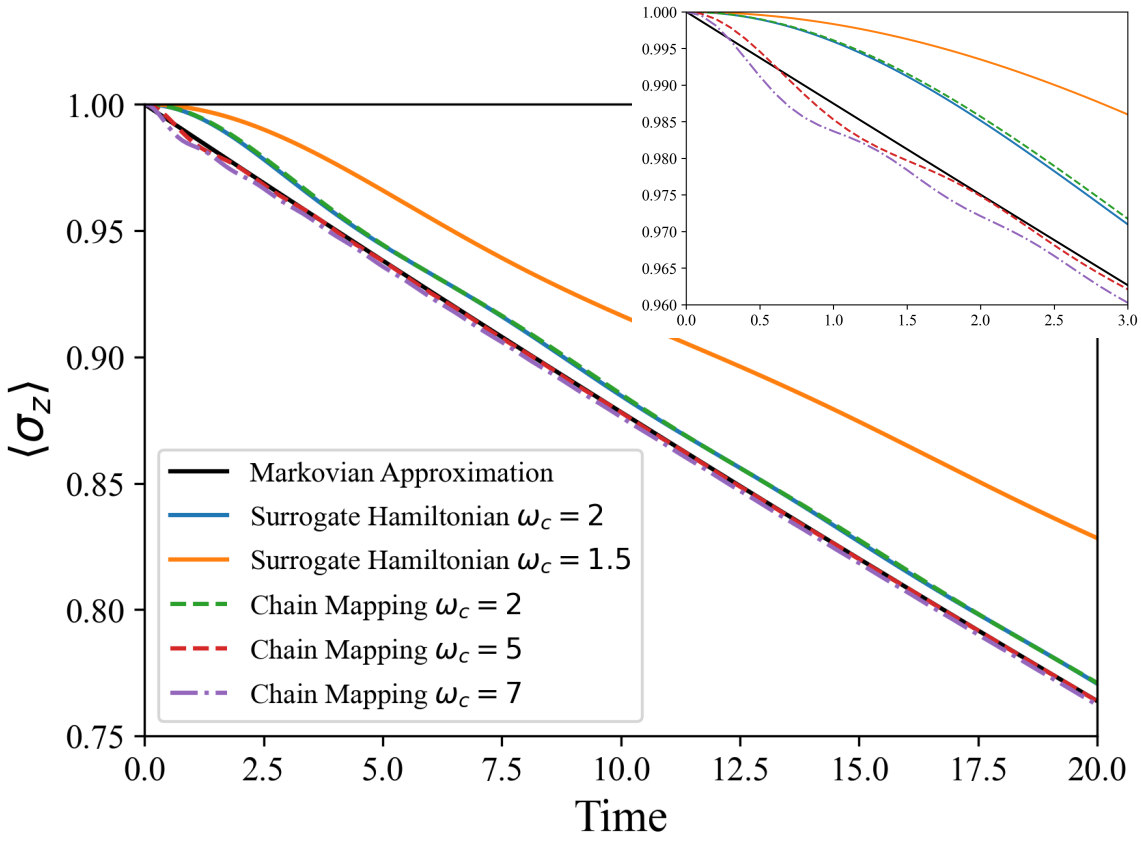



Figure 4.2: Surrogate hamiltonian and chain mapping approximation of the spin-boson model compared to the Markov approximation (black) for weak coupling ($\alpha = 0.001$). The Markov approximation is relatively unaffected by the cut-off frequency, with both $\omega_c = 2$ and $\omega_c = 5$ represented by the black line. The chain mapping approach with $\omega_c = 5$ (orange) is the closest to the Markov approximation over short times closely followed by $\omega_c = 7$, but at very short times (inlay) the approximations exhibit is a small deviation from the dynamics predicted by the Markov approximation. This occurs because at very short times there are built up correlations that are not captured by the Markov approximation [41, 44]. The

| **Method** | $\omega_c$ | **Average Gradient** |
|---|---|---|
| Markovian Approximation | 2/5/7 | $-0.011$ |
| Surrogate Hamiltonian | 1.5 | $-0.008$ |
| | 2 | $-0.014$ |
| Chain Mapping | 2 | $-0.011$ |
| | 5 | $-0.011$ |
| | 7 | $-0.011$ |

Table 4.1: Average decay rate over the time interval t=10 to t=20 in figure 4.2 for different methods and cut-off frequencies $\omega_c$.

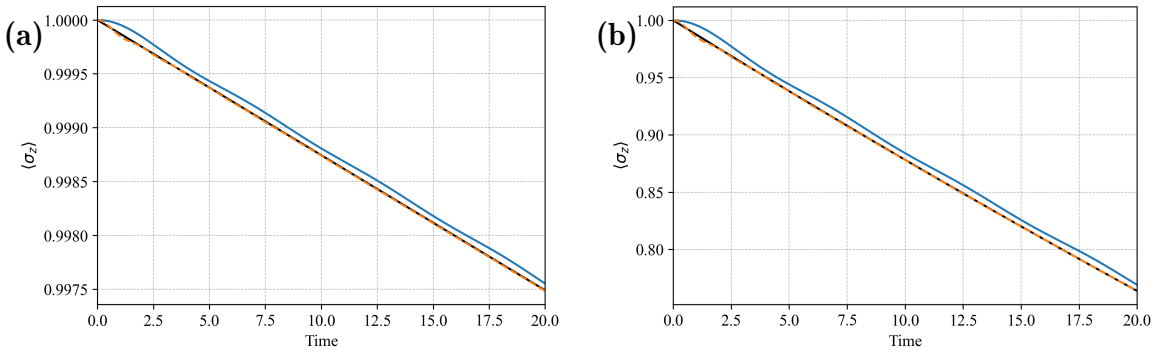



Figure 4.3: Looking at the performance of the surrogate hamiltonian approximation (blue) with $\omega_c = 2$ and chain mapping approximation (orange) with $\omega_c = 5$ compared to the Markov approximation (black) with $\omega_c = 2$. for weak coupling (a)$\alpha = 10^{-5}$, (b) $\alpha = 0.001$, the chain mapping approximation performs better for this system. The slight difference in the approximation from the surrogate hamiltonian is likely due to the small $\omega_0 = 1$. This limits the cut off frequency required for self convergence to $\omega_c = 2$ which is too small to capture the dynamics of the environment even at short times.

## 4.2 Strong Coupling regime

In the strong coupling regime ($\alpha \geq 0.1$), the assumptions underpinning the Markov approximation break down entirely. Non-Markovian effects become dominant, and larger correlations between system and environment significantly alter the dynamics. In this regime, approximate methods like the Surrogate Hamiltonian and Chain Mapping must independently capture genuine non-Markovian behaviour, without the support of a reliable analytical benchmark.

Figure 4.4 shows the system's dynamics in this regime by plotting the evolution of $\langle \sigma_z(t) \rangle$ for $\alpha = 0.1$ across a range of cut-off frequencies. Several features distinguish these results from the predictions of the Markov approximation. Firstly, the system no longer relaxes to zero but instead settles at a finite equilibrium value, indicating incomplete decoherence. Secondly, the relaxation process exhibits damped oscillations rather than a smooth exponential decay, reflecting persistent system-environment correlations. Finally, both the oscillation patterns and steady-state values are highly sensitive to the bath cut-off frequency $\omega_c$, reflecting how a broader bath spectrum influences the energy exchange between system and environment.

Figure 4.4 shows the evolution of $\langle \sigma_z(t) \rangle$ for $\alpha = 0.1$ at various cut-off frequencies. The dynamics differ dramatically from Markovian predictions in several key aspects. First, the system no longer decays to zero but reaches a finite steady-state value, indicating incomplete decoherence. Second, the approach to steady state exhibits pronounced oscillations rather than smooth exponential decay. Third, both the steady-state value and oscillation patterns depend strongly on the cut-off frequency $\omega_c$. This dependence is caused by the equilibrium state having more contributions from both the system and the bath when the coupling strength is higher. This increased influence from the bath means that when the cut-off frequency is higher, the equilibrium energy is also higher. This is a physical implication of cut-off frequency.

The Chain Mapping method successfully captures these non-Markovian features across all tested cut-off frequencies. For $\omega_c = 2$, we observe long-lived coherent oscillations and a lower equilibrium energy. This is similar to the approximated dynamics provided by the surrogate Hamiltonian, with only mild disagreements in oscillatory behaviour.

As $\omega_c$ increases to 5, 7 and 10, the oscillations become more irregular and damp more quickly, reflecting the participation of a broader range of bath frequencies in the dynamics. As seen in Figure 4.4, this oscillatory behaviour causes increases in $\langle\sigma_z(t)\rangle$ indicative of state space volume increases overall. This confirms the emergence of non-Markovian dynamics captured by the chain mapping method at higher cut-off frequencies. The relaxation dynamics predicted in the strong coupling regime by these methods deviates from the long term exponential decay given by the Markov approximation. This is expected as the Markov approximation is unable to capture these non-Markovian effects.

The Surrogate Hamiltonian method, however, is more limited in this regime. Its frequency sampling constraint $\omega_c \leq 2\omega_0$ prevents it from exploring the broader spectral range required to resolve strong coupling effects. This highlights a key limitation: accurately capturing strong coupling dynamics requires representing a wider frequency spectrum than the Surrogate Hamiltonian can accommodate under its current symmetric sampling strategy. Further investigation is needed to confirm other sampling strategies that still ensure self convergence.

Together, these results confirm that strong coupling pushes the system into a truly non-Markovian regime where approximations that eliminate environmental memory fail. The observed deviations from exponential decay and increases in state space volume (as seen in Figure 4.4) underscore the presence of information backflow and long-lived correlations—features that are correctly captured only when the bath is modelled in sufficient detail.

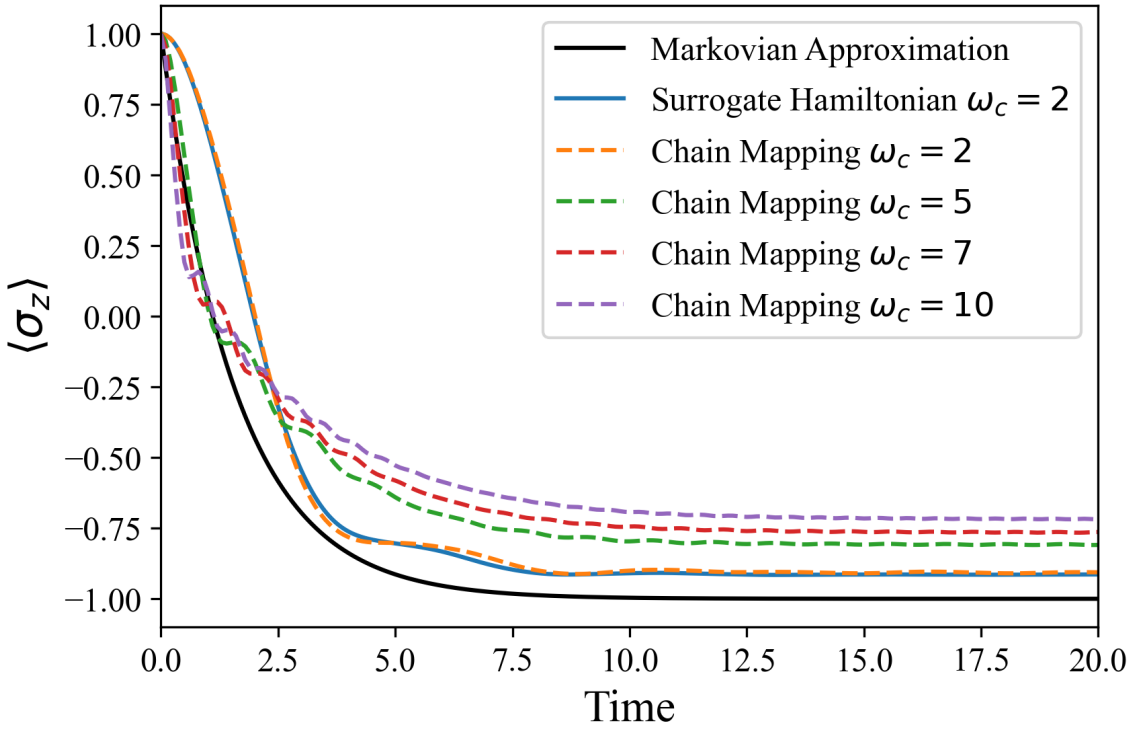



Figure 4.4: In the strong coupling regime $\alpha = 0.1$, the Markov Approximation (black) is no longer valid. Here, both the surrogate hamiltonian approximation (blue) and the chain mapping approximation (orange) do not show relaxation to the $|\downarrow\rangle$ state. For (a)$\omega_c = 2$, the chain mapping approximation is very similar to the surrogate Hamiltonian approximation, but for (b) $\omega_c \geq 5$ the chain mapping approximation relaxes quite differently.

# Chapter 5

# Conclusion and Outlook

The accurate simulation of open quantum systems remains one of the central challenges in quantum physics, as it requires methods that can efficiently capture the infinite degrees of freedom associated with environmental baths in some finite way whilst preserving key non-Markovian features. This thesis has addressed this challenge by implementing and comparing two effective mode approaches: chain mapping, which preserves the exact spectral function through unitary transformation, and the surrogate Hamiltonian method, which approximates the bath through direct mode sampling. Both techniques transform a continuous bath into a discrete, computationally manageable representation. Using numerical simulations of the spin-boson model, we have examined the convergence behaviour, parameter sensitivity, and limitations of each method across a range of regimes exhibiting both Markovian and non-Markovian dynamics.

While both approaches rely on truncating the environment and propagating the system and bath together, they differ fundamentally in how this truncation is performed. The chain mapping method transforms the original bath into a one-dimensional chain with nearest-neighbour interactions via an exact mapping. The resulting chain modes do not correspond directly to physical bath modes. Rather, they encode the influence of the full environment on the system in a compressed, structured form. Whilst robust, this method is complicated to apply to new models as it requires the derivation of specific orthogonal polynomials for the transformation operator.

In contrast, the Surrogate Hamiltonian approach directly samples a finite subset of physical bath modes. This procedure is more physically intuitive than chain mapping, but the sampling must satisfy strict criteria, such as mode symmetry and proportional density of states to ensure self-convergence. While this flexibility allows for targeted modelling, it also introduces greater potential for user error and imposes geometric constraints that can restrict the accessible parameter space.

Despite their differing mathematical foundations, both methods ultimately aim to represent the same underlying system-environment dynamics. A central focus of this thesis was to compare their performance across a range of regimes, with particular attention to their ability to capture non-Markovian features like correlations. To this end, we presented numerical simulations of the spin-boson model, identifying the conditions under which the methods agree, diverge, and how their accuracy depends on

parameters such as coupling strength and bath cut-off frequency. For the chain mapping approach, systems with larger cut-off frequencies tended to give more predictable results in both the weak and strong coupling regime.

The results highlight a strong sensitivity to the bath cut-off frequency in both approaches. This sensitivity stems partly from the use of a hard cut-off spectral density, but also from the fact that the structure of the bath is explicitly included in the dynamics. Even at weak coupling, subtle differences in how the environment is sampled can lead to noticeable deviations in system evolution.

In the strong coupling regime, the influence of the cut-off frequency becomes even more pronounced. Not only does the sampling procedure itself play a critical role, but increased system-bath entanglement leads to stronger information backflow and long-lived memory effects. These dynamics are particularly sensitive to the distribution and extent of bath modes included in the full model.

Overall, the results show that the Chain Mapping method is well-suited for capturing long-time dynamics and strong system-bath correlations, especially in highly non-Markovian regimes. However, this method is quite complex and may be difficult to apply to some systems. In contrast, the Surrogate Hamiltonian provides a computationally efficient alternative for short-time and weak coupling scenarios, albeit with stricter limitations on spectral range and convergence criteria. The simple construction allows for easy simulation of short term dynamics. By characterising the convergence behaviour and practical constraints of both methods, this thesis offers a foundation for choosing appropriate modelling strategies in future studies of open quantum systems.

Taken together, these findings point toward a broader outlook. Effective mode approaches are not only valuable computational tools, but also provide insight into how environment structure and coupling influence quantum coherence and memory effects. As quantum technologies continue to evolve, the ability to model realistic, structured environments will be essential. The methods explored in this thesis offer an insight into such methods and clarify which directions future research could lead to.

A natural direction for future research lies in the development of alternative sampling strategies for the Surrogate Hamiltonian approach. The current strategy, which relies on symmetric and equidistant frequency sampling around the system frequency, imposes geometric constraints that restrict its applicability, particularly when modelling environments with broad or highly structured spectral densities. Designing adaptive or optimised sampling schemes that relax these constraints while preserving numerical convergence could significantly expand the method's utility across a wider class of physical systems. Additionally, more comprehensive studies of the spin-boson model, especially under different initial states and observables, would help clarify how each method's bath truncation affects the system dynamics and treatment of coherences.

Another promising avenue for future work is the application of effective mode methods to larger and more complex open quantum systems. Extending these techniques to two-qubit systems could shed light on multipartite entanglement and help clarify the

relationship between global and local master equations. While preliminary work has applied chain mapping to such models [24], further exploration is needed to understand how correlations are preserved and distributed across the effective bath, and how these methods compare in accuracy and efficiency when scaling to more qubits or coupled systems.

In conclusion, this thesis has provided a detailed comparison of two effective mode approaches for simulating open quantum systems, highlighting their strengths, limitations, and areas of applicability. Chain mapping reveals how environmental memory emerges from hierarchical correlation structures, while the surrogate Hamiltonian method clearly shows the dependence on frequency sampling. Together, they provide complementary perspectives on the approximation of open quantum systems.

# Appendix A

# Numerics

## A.1 Parameters of the calculations

Unless stated otherwise, all parameters are in atomic units and T=0.

### Chapter 2 parameters

Table A.1: Chain mapping model parameters

| **Parameter** | **Value** |
|---|---|
| Bath mode dimension ($d$) | 3 |
| System frequency ($\omega_0$) | 1.0 |
| Ohmic parameter ($s$) | 1.0 |

Table A.2: Chain mapping TEBD engine parameters

| **Parameter** | **Value** |
|---|---|
| Number of steps between measurement ($N_{\text{steps}}$) | 1 |
| Time step size ($\Delta t$) | 0.1 |
| Suzuki-Trotter decomposition order | 4 |
| SVD truncation threshold | $1 \times 10^{-12}$ |

## Chapter 3 parameters

Table A.3: Surrogate Hamiltonian parameters

| **Parameter** | **Value** |
|---|---|
| System Frequency ($\omega_0$) | 1.0 |
| Ohmic parameter ($s$) | 1.0 |
| Bath mode dimension ($N_{\text{levels}}$) | 3 |

## Chapter 4 parameters

Table A.4: Chain mapping model parameters

| **Parameter** | **Value** |
|---|---|
| Chain length ($L$) | 100 |
| Bath mode dimension ($d$) | 3 |
| System frequency ($\omega_0$) | 1.0 |
| Ohmic parameter ($s$) | 1.0 |

Table A.5: Chain mapping TEBD engine parameters

| **Parameter** | **Value** |
|---|---|
| Number of steps between measurement ($N_{\text{steps}}$) | 1 |
| Time step size ($\Delta t$) | 0.1 |
| Suzuki-Trotter decomposition order | 4 |
| SVD truncation threshold | $1 \times 10^{-12}$ |
| Maximum bond dimension ($\chi_{\max}$) | 50 |
| SVD truncation threshold | $1 \times 10^{-12}$ |

Table A.6: Surrogate Hamiltonian parameters

| **Parameter** | **Value** |
|---|---|
| System frequency ($\omega_0$) | 1.0 |
| Ohmic parameter ($s$) | 1.0 |
| Number of bath modes ($N$) | 15 |
| Bath mode dimension ($N_{\text{levels}}$) | 2 |

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